

Channel-Level Closure of Four Minimal Einstein–Cartan–Holst Dark-Energy Routes and Perturbation Transparency for Scalar Matter

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We assess four enumerated minimal-Einstein-Cartan-Holst (ECH) spin-torsion channels as candidate sources of late-time dark energy and find that each fails at the amplitude level under stated assumptions. This is a channel-level assessment, *not* an operator-level theorem: the four enumerated routes (NJL, one-loop EA, Immirzi running, parity-CMB) are not proven to be a complete diffeomorphism-invariant operator basis for the minimal-ECH effective action; we acknowledge missing operators (Jackiw-Pi gravitational Chern-Simons $R \wedge \tilde{R}$, parity-odd four-fermion partner with $\gamma_{\text{BI}}/(\gamma_{\text{BI}}^2 + 1) \cdot 8\pi G$ coefficient) explicitly in Sec. IV and Sec. XI. The dark-energy mapping rests on a phenomenological on-shell scaling ansatz whose off-shell mass dimension is +1 rather than +4 (Appendix B); we treat this scaling explicitly as an ansatz, not a derivation. Through 7 foundation studies (Foundations A–G) and 6 observational research branches (Branches H, J, L, M, N, O) we report 13 logically-independent mechanism-class constraints that collectively constrain the enumerated channels of the minimal-ECH route from the quantum bounce to observable dark energy. The central result is a perturbation-transparency theorem: for canonical scalar matter, torsion vanishes at all perturbation orders, the Holst dual contraction $\epsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}$ vanishes identically on the Levi-Civita connection ($T = 0$) by the first (algebraic) Bianchi identity $R_{\mu[\nu\rho\sigma]} = 0$, and the Holst sector therefore decouples from all scalar/tensor perturbation equations of motion (Sec. X).^a The proposed link from ECH to a late-time vacuum energy requires a phenomenological dimensional ansatz beyond the minimal framework (Appendix B), and is constrained at the channel-amplitude level by 13 logically-independent barriers (Sec. IX; 14 historical catalog entries, of which B8 is subsumed by B14 per the perturbation-transparency result). A structural tension (Sec. XIV D) exists between the dark-energy mechanism, which requires $N_{\text{tot}} \approx 92$ post-bounce e -folds, and the matter-bounce $f_{\text{NL}} = -35/8$ signature, which would be *definitively* erased at SPHEREx-accessible comoving wavenumbers by that many e -folds (a contracting-phase quantity mode with $k_{\text{SPHEREx}} \sim 10^{-1} h/\text{Mpc}$ is pushed to $k_{\text{bounce}}^{\text{phys}} \sim k_{\text{SPHEREx}}^{\text{phys}} e^{N_{\text{tot}} - N_{\text{exit}}} \sim e^{32} k_{\text{SPHEREx}}^{\text{phys}}$ at $N_{\text{tot}} \sim 92$, $N_{\text{exit}} \sim 60$ (the relative e -fold differential between bounce and CMB horizon-exit; comoving wavenumbers k are constant by definition and only physical scales scale with $a^{-1} \propto e^{-N}$), deep inside the inflationary subhorizon regime carrying purely vacuum-inflationary fluctuations rather than matter-bounce contraction modes); the minimal-ECH four-route channel set is therefore tightly constrained as both a dark-energy generator and a matter-bounce host. The two predictions discussed below as “surviving” are accordingly *not* predictions of ECH itself, but bounce-class and GR+ALP-class observables that the channel-level assessment does not forbid in the parameter regimes complementary to its dark-energy mechanism; we report them here because they remain testable signatures of any bounce model in the broader programme. Specifically: (i) $f_{\text{NL}} = -35/8$ is a property of the *matter-bounce class* [1], derived from the contraction-phase cubic action with no ECH input (a detailed multi-tracer SPHEREx Fisher forecast is presented in a companion work in preparation [2]); and (ii) spectator-ALP birefringence $\beta \approx 0.27^\circ$ is a *benchmark consistency point*, not an ECH prediction: it sits inside the WMAP+Planck 1σ band $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$ ($\sim 3.6\sigma$ from $\beta = 0$, first reported by Minami & Komatsu [3] and refined by Eskilt & Komatsu [4]), and is comparable to the independent ACT DR6 follow-up $\beta = 0.215^\circ \pm 0.074^\circ$ at $\sim 2.9\sigma$ (Diego-Palazuelos & Komatsu [5]); the same benchmark arises in any GR+ALP setup with the same parameters and is *not* derived from the ECH action. The role of this paper is the *channel-level closure* of the four enumerated minimal-ECH dark-energy routes (Sec. IV) at amplitude-budget granularity; we do not claim a full operator-basis closure of the most general ECH parity-odd / torsion-sourced sector (Jackiw–Pi gravitational Chern–Simons and the parity-odd four-fermion partner of R1 are listed but not separately enumerated in the four routes; their explicit closure is left to a follow-up operator-level analysis). The complementary observational programme is hosted in companion works in preparation [2, 6]. $\Lambda\text{CDM} + \Delta N_{\text{eff}}$ MCMC verification, NaMaster pipeline validation, and ALP parameter fitting are documented separately in companion work in preparation [6].

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^a This Bianchi-identity vanishing is distinct from — and should

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not be confused with — the Pontryagin density $\propto R \tilde{R} = \epsilon^{\mu\nu\rho\sigma} R_{\mu\nu}{}^{\alpha\beta} R_{\rho\sigma\alpha\beta}$, which involves *two* curvature tensors and is a separate topological invariant. The Holst dual contraction has only *one* curvature. In differential-form language $e^I \wedge e^J \wedge R_{IJ} = -NY + T^I \wedge T_I$ (Nieh–Yan density plus torsion-squared); both pieces vanish at $T = 0$, recovering the Bianchi-vanishing result. Earlier versions of this manuscript erroneously identified the two; the correction preserves the headline conclusion (Holst sector decouples from scalar/tensor EOM) by a Bianchi-trivial argument rather than a Pontryagin total-derivative argument.

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I. INTRODUCTION

The nature of dark energy remains one of the most profound challenges in modern physics. While the Λ CDM model successfully accounts for observed cosmic acceleration [7], it faces severe theoretical difficulties—most notably the cosmological constant problem [8]. DESI 2024–2025 BAO results suggest dynamical dark energy at $3.1\text{--}4.2\sigma$ (dataset-dependent) [9, 10], adding urgency to the search for extensions of the standard model.

This work presents and closes a phenomenological framework that connects ECH spin-torsion cosmology to dark energy. The structural conclusion is a *channel-level amplitude closure* of the four enumerated minimal-ECH dark-energy routes (Sec. IV): under the stated assumptions, the 14 constraints (Sec. IX, 13 logically-independent with B8 subsumed by B14) close those routes at amplitude-budget granularity. The surviving phenomenological predictors—matter-bounce f_{NL} and spectator-ALP birefringence—are mechanism-independent and shared by other UV completions.

a. Scope and limitations. The four routes assessed below are minimal channels rather than a complete diffeomorphism-invariant operator basis. In particular, the Jackiw–Pi gravitational Chern–Simons term $R\tilde{A}$ and the parity-odd four-fermion partner of Route 1 (carrying coefficient $\gamma_{\text{BI}}/(\gamma_{\text{BI}}^2+1)\cdot 8\pi G$) are excluded from the enumeration and their explicit closure is left to a follow-up operator-basis analysis. The dark-energy mapping presented in Sec. II C and Appendix B rests on a phenomenological on-shell scaling ansatz: the leading parity-odd operator written in Eq. (6) has off-shell mass dimension +1 and acquires its ρ_Λ mapping only through on-shell evaluation at Planck-scale bounce densities; we treat this mapping explicitly as an ansatz, not a derivation. The perturbation-transparency result of Sec. X is restricted to canonical scalar matter: fermion spin density, propagating torsion (Poincaré gauge theory), dynamical Immirzi fields, non-minimal matter couplings, and boundary/topological sectors are explicitly outside its scope. Within those caveats, the four channels are assessed at the amplitude-budget granularity at which observations discriminate, and the resulting closure is a channel-level

statement under specified assumptions rather than a full operator-level theorem.

Our approach builds on three theoretical pillars:

1. *Loop Quantum Cosmology* (LQC), providing a non-singular quantum bounce replacing the classical Big Bang singularity [11]. The bounce occurs at $\rho_{\text{crit}} \simeq 0.27\text{--}0.41 \rho_{\text{Pl}}$ (Barbero-Immirzi entropy-counting scheme dependent; Sec. IIB).
2. *Einstein-Cartan theory* incorporating fermionic spin-torsion coupling, generating four-fermion contact interactions and preventing gravitational singularities through torsion-induced repulsion at extreme densities [12, 13].
3. *Black hole universe origin*, where a rotating parent black hole spawns a non-singular baby universe through torsion-regulated gravitational collapse [14]. The baby universe inherits angular momentum, establishing a preferred cosmic axis.

A. Theoretical Foundations and Novel Synthesis

Our framework collects well-established theoretical components and tests them as a channel-level amplitude closure of the four enumerated minimal-ECH dark-energy routes (we do not claim a full operator-basis closure; see Sec. IV “Scope” for the explicit operators omitted). The original contributions are:

1. *14-constraint catalog and perturbation-transparency observation:* Through 7 foundation studies (Foundations A–G) and 6 observational research branches (Branches H, J, L, M, N, O), we establish 14 mechanism-class structural constraints (one of which, B8, is the observational consequence of the perturbation-transparency theorem B14 and is retained in the catalog for historical mechanism-class completeness) mapping the minimal ECH parameter space. The central result is that minimal ECH gravity is perturbation-transparent for canonical scalar matter: torsion vanishes at all orders, the Holst sector decouples cleanly from scalar/tensor observables, and tests of γ shift to nonperturbative parity-violating channels (ALP birefringence, primordial GWs).
2. *Structural tension between dark-energy suppression and bounce f_{NL} :* We identify an incompatibility between the inflationary-suppression dark-energy mechanism ($N_{\text{tot}} \approx 92$ e -folds required) and the matter-bounce $f_{\text{NL}} = -35/8$ prediction (*definitively* erased by $N_{\text{tot}} \gtrsim 60$ inflationary e -folds at SPHEREx-relevant scales, since the SPHEREx accessible band $k \sim 10^{-4}\text{--}10^{-1} h/\text{Mpc}$ maps to bounce-era *physical* scales $k_{\text{bounce}}^{\text{phys}} = k_{\text{SPHEREx}}^{\text{phys}} e^{N_{\text{tot}} - N_{\text{exit}}}$ at $N_{\text{tot}} \sim 92$ and $N_{\text{exit}} \sim 60$ (relative e -fold differential ~ 32 ; comoving k are constant by definition, only physical scales

TABLE I. Executive summary: systematic investigation of bounce-cosmology dark-energy routes within ECH. Structural constraints narrow phenomenological pathways; the surviving testable prediction is the matter-bounce $f_{\text{NL}} = -35/8$ [1].

Question	Result	Status
Can bounce derive dark energy?	14 constraints map minimal-ECH route space	Phen. assumption ^a required.
Is there a nonsingular bounce?	LQC: $\rho_c \simeq 0.27\text{--}0.41 \rho_{\text{Pl}}$	Yes (LQC holonomy).
ECH visible in scalar/tensor pert.?	Perturbation-transparency result	Decouples; tests $\rightarrow$ ALP/GW.
Testable prediction?	$f_{\text{NL}} = -35/8$ (Paper II forecast ^b)	Yes , class-level ^c .
Mechanism-independence?	Class-level: scalar-only $w = 0$ matter-bounce ^c	Not a distinctive ECH prediction.
H_0/σ_8 tension resolution?	$H_0 = 67.68 \pm 1.06$, $\Delta N_{\text{eff}} \approx 0$	Recovers Λ CDM.

^aReparameterized as sensitivity to N_{tot} ; not solved. ^b3–5 σ realistic after full systematic budget (GR-projection, b_ϕ uncertainty, photo- z degradation) under Heinrich+2024 $\sigma(f_{\text{NL}}) \approx 0.7$ — detailed Fisher forecast in companion work in preparation [2]. ^cClass-level: scalar-only $w = 0$ matter-bounce under Assumption (f) of the companion forecast [2]; not fully mechanism-independent across the bouncing-cosmology landscape; not a distinctive ECH prediction. MCMC details in companion work in preparation [6].

scale with $a^{-1} \propto e^{-N}$) that lie deep inside the inflationary subhorizon regime where the observable bispectrum is dominated by vacuum-inflationary modes rather than the matter-bounce contraction modes; see Sec. XIV D), establishing that these are independent observational programs.

3. *Survival of mechanism-independent tests:* Despite ECH structural closure, two predictions of the broader bounce/ALP landscape survive and are testable by SPHEREx (2028) and LiteBIRD (early 2030s) independently of any specific UV completion.

B. Paper Organization

Section II develops the ECH theoretical framework. Section III presents observational signatures (galaxy spin null, CMB EB). Section IV closes each of the four standard ECH routes (NJL, one-loop EA, Immirzi running, parity-CMB) at the amplitude level. Section V covers the galaxy-spin data methods. Sections VI–VIII provide systematic analysis, falsification criteria, and related work. The core results occupy Secs. IX–XI: the 13 logically-independent structural barriers (14 historical catalog entries), perturbation-transparency observation, and hybrid loophole rejection. Section XII discusses the inflationary suppression factor and theoretical implications. Section XIII summarizes surviving mechanism-independent tests. Section XIV addresses limitations. Section XV concludes.

Companion paper.— Λ CDM+ ΔN_{eff} MCMC verification (Cobaya v3.6.1, **309,189** frozen accepted samples across two converged dataset combinations: 176,240 full-tension + 132,949 Planck+BAO+SN; see Paper I(b) [6] Table I for the per-dataset breakdown), NaMaster pseudo- C_ℓ pipeline validation, spectator-ALP MCMC parameter fitting, and a cross-paper reproducibility manifest are reported in Paper I(b) [6]. Cosmological parameter values referenced in this paper ($H_0 = 67.68 \pm 1.06$, $\Delta N_{\text{eff}} \approx 0$, etc.) are drawn from the companion internal MCMC analysis (Paper I(b) [6], *in preparation*); they are

documented internally rather than as externally citable arXiv-posted numbers, and should be read as internal-analysis inputs to the present structural argument rather than as independently peer-reviewable values until Paper I(b) is publicly posted.

Appendices provide the parameter summary (Appendix A) and dimensional analysis (Appendix B). Supplementary materials are at <https://github.com/Hubify-Projects/bigbounce>.

II. THEORETICAL FRAMEWORK

A. Loop Quantum Cosmology and the Holst Action

1. Einstein-Cartan-Holst Action

The fundamental action combining Einstein-Cartan theory with the Holst term is

$$S_{\text{ECH}} = \frac{1}{16\pi G} \int d^4x e \left[e_a^\mu e_b^\nu R_{\mu\nu}^{ab} + \frac{1}{\gamma} \varepsilon^{abcd} e_a^\mu e_b^\nu R_{cd\mu\nu} + \frac{1}{4} T^{abc} T_{abc} \right] + S_{\text{matter}}, \quad (1)$$

where $e = \det(e_\mu^a)$ is the tetrad determinant, $R_{\mu\nu}^{ab}$ is the curvature of the Lorentz connection, γ is the Barbero-Immirzi parameter, and T^{abc} is the torsion tensor. The Holst term contributes non-trivially when fermions are present, as established by Freidel, Minic & Takeuchi [15], who showed that the Barbero-Immirzi parameter becomes physically observable through its coupling to fermionic matter; the explicit γ -dependence it induces appears in the four-fermion contact term, Eq. (4). The $T^{abc}T_{abc}$ term in Eq. (1) is a shorthand for the four-fermion contact interaction obtained after integrating out the non-propagating torsion; it is not an independently specified kinetic term and is *not* varied independently — the connection variation is performed on the Einstein–Cartan–Holst+Dirac action alone, with Eq. (3) the resulting Cartan equation, so no double counting arises.

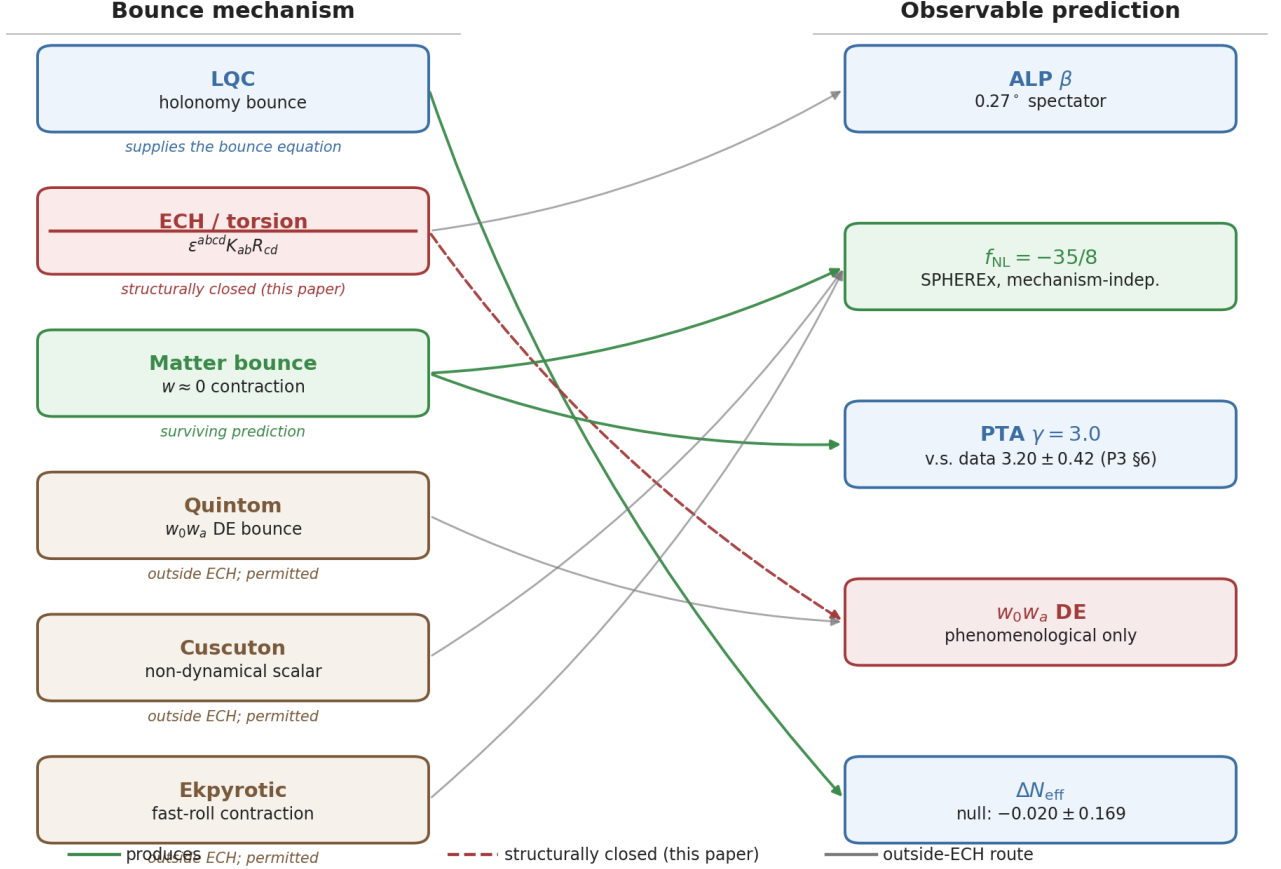


FIG. 1. Bounce-mechanism $\rightarrow$ observable-prediction map. Left column: candidate non-singular bounce mechanisms (LQC, ECH/torsion, matter bounce, quintom-B, Cuscuton, ekpyrotic). Right column: distinctive observable channels. ECH appears bordered with a dashed box marked *channel-level closure under stated assumptions (this paper)*—the 14-constraint catalog narrows the four enumerated minimal-ECH dark-energy channels to zero phenomenologically free pathways within those channels. The burned-in PTA annotation $\gamma = 3.20 \pm 0.42$ is the superseded pre-real-KDE draft value; the current real-KDE reanalysis gives $\gamma = 2.567 \pm 0.382$ (Sec. X G), and the figure annotation is pending regeneration.

The Barbero-Immirzi parameter is fixed by LQG black hole entropy in the SU(2) full-counting scheme used here:

$$\gamma_{\text{SU}(2)} \approx 0.274, \quad (2)$$

where the apparent uncertainty range is *scheme dependence rather than a statistical or theoretical error*: the U(1) horizon-state counting [16] gives $\gamma_{\text{U}(1)} \approx 0.127$ (using $\gamma = \ln 2 / (\pi \sqrt{3})$), the refined SU(2) full counting [17, 18] gives $\gamma_{\text{SU}(2)} \approx 0.274$ (adopted in this paper), and the further Domagała–Lewandowski–Meissner refinement gives $\gamma_{\text{DLM}} \approx 0.2375$. Domagała–Lewandowski and Meissner do *not* quote a ± 0.020 statistical uncertainty; the ~ 0.020 figure that appears in the parameter-budget table (Appendix B) is the spread between counting prescriptions, retained as an effective range only and *not* propagated as a statistical error in any quantitative claim.

2. Derivation of the Parity-Odd Term

Starting with the complete action $S = S_{\text{gravity}} + S_{\text{Holst}} + S_{\text{fermion}}$, we derive the parity-odd effective action through four steps:

Step 1: Torsion Activation.—Torsion is determined algebraically by the fermionic spin density:

$$T^{abc} = 8\pi G S^{abc}, \quad (3)$$

where $S^{abc} = \frac{1}{4} \bar{\psi} \gamma^{[a} \gamma^{bc]} \psi$.¹

¹ Single-convention derivation (fixing the paper’s normalization once): we use the form-language torsion definition $T^a \equiv De^a$ (components $T^\lambda_{\mu\nu} = 2\Gamma^\lambda_{[\mu\nu]}$) and the Hermitian (symmetrized) minimally coupled Dirac action. Varying that action with respect to the connection gives the spin current $S^{abc} \equiv \delta \mathcal{L}_\psi / \delta \omega_{abc} = \frac{1}{4} \bar{\psi} \gamma^{[a} \gamma^{bc]} \psi = \frac{1}{4} \epsilon^{abcd} J^5_d$ — *totally antisymmetric*.

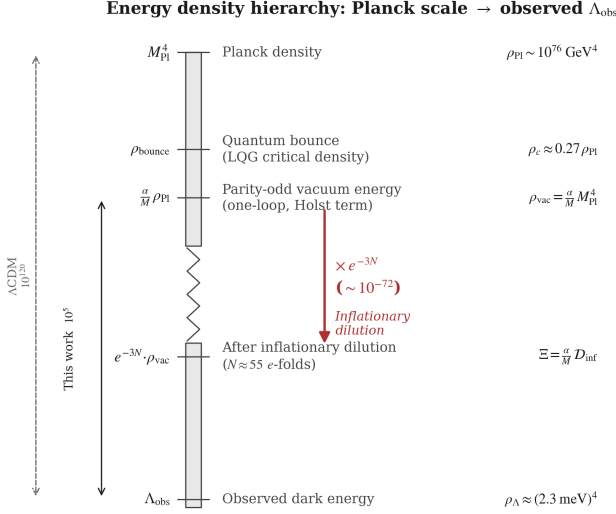


FIG. 2. Energy density hierarchy from the Planck scale to the observed dark energy scale, illustrating the phenomenological scaling ansatz $\rho_{\text{vac}} \sim [(\alpha/M) M_{\text{Pl}}] M_{\text{Pl}}^4$ (Sec. II A 2, Appendix B). This ansatz is dimensionally correct on-shell at the bounce but is *not* derived from the ECH action. The panel’s burned-in dilution waypoint is an *illustrative* $N \approx 55$, $e^{-3N} \sim 10^{-72}$ schematic; the quantitative book-keeping of Sec. XII A and Appendix B uses $N_{\text{tot}} \approx 92$ with $D_{\text{inf}} \sim 10^{-121}$.

Step 2: Four-Fermion Contact Interaction.—Substituting Eq. (3) and integrating out torsion:

$$\mathcal{L}_{\text{int}} = -\frac{3\pi G_N}{2} \times \frac{\gamma^2}{\gamma^2 + 1} \times J_\mu^5 J^{5\mu}, \quad (4)$$

where $J^{5\mu} = \bar{\psi} \gamma^\mu \gamma^5 \psi$ is the axial current.

Step 3: Parity-Odd Effective Action.—Motivated by the Holst+non-minimal-fermion construction of Mercuri [19] (which shows that the Nieh–Yan invariant is

ric, so all torsion trace parts vanish and the Cartan equation is exactly $T^{abc} = \kappa S^{abc}$, Eq. (3). Consistency proof by back-substitution: $S_{abc} S^{abc} = \frac{1}{16} \epsilon_{abcd} \epsilon^{abce} J_e^{5d} J_e^5 = -\frac{3}{8} J_\mu^5 J^{5\mu}$ (Lorentzian $\epsilon_{abcd} \epsilon^{abce} = -3! \delta_d^e$, with mostly-plus signature $g = \text{diag}(-, +, +, +)$ and $\epsilon^{0123} = +1$); explicitly, the on-shell substitution $T^{abc} = \kappa S^{abc}$ gives $\frac{1}{4} T^{abc} T_{abc} = \frac{\kappa^2}{4} S^{abc} S_{abc}$, and the standard Hehl–Datta bookkeeping (the gravitational contact piece plus the fermionic spin–torsion coupling; the same sign manipulation quoted at Eq. (13)) nets to $\mathcal{L}_{\text{int}} = +\frac{\kappa}{2} S_{abc} S^{abc}$, so integrating out torsion yields $\mathcal{L}_{\text{int}} = -\frac{3\kappa}{16} J_\mu^5 J^{5\mu}$ —precisely the $\gamma \rightarrow \infty$ (pure Einstein–Cartan) limit of Eq. (4) and the Hehl–Datta coefficient of Eq. (13), closing the chain with no residual factor. The $T^{abc} = (\kappa/2) \bar{\psi} \gamma^{[a} \gamma^b \gamma^c] \psi$ form quoted in the original Hehl–Datta-era literature (Sec. IV A) uses the half-weight torsion definition $T^\lambda_{\mu\nu} = \Gamma^\lambda_{[\mu\nu]}$ and maps to Eq. (3) exactly (the full-weight definition $T^\lambda_{\mu\nu} = 2\Gamma^\lambda_{[\mu\nu]}$ used here absorbs the factor of 2, so the half-weight $(\kappa/2)$ prefactor and the full-weight κ prefactor describe the same torsion field); the physical contact term $-\frac{3\kappa}{16} (J^5)^2$ is identical in both conventions.

reconstructed and the Barbero–Immirzi parameter drops out of the classical dynamics), we introduce as a phenomenological ansatz the parity-odd term:

$$S_{\text{eff}} = \frac{\alpha}{M} \int e_I \wedge e_J \wedge \mathcal{F}^{IJ}[K, \hat{R}], \quad (5)$$

where $M = M_{\text{area-gap}} \sim M_{\text{Pl}}/\sqrt{\gamma}$ is the LQG area-gap mass scale (from the LQG area-gap $\Delta \propto \gamma \ell_P^2$, the inverse-length / mass scale is $M_\Delta \sim M_{\text{Pl}}/\sqrt{\gamma}$ up to numerical constants), α is a dimensionless coupling, and $\mathcal{F}^{IJ}[K, \hat{R}]$ denotes the curvature two-form of the full (torsionful) connection, written as a functional of the contorsion K and the Levi-Civita curvature $\hat{R}$; the component form Eq. (6) displays its leading contribution. (The calligraphic $\mathcal{F}$ is reserved for this gravitational curvature; the electromagnetic field strength of Sec. IV D is written $F_{\mu\nu}$.)

In components, the leading contribution reduces to

$$S_{\text{eff}} = \int d^4x \sqrt{-g} \frac{\alpha}{M} \varepsilon^{\mu\nu\rho\sigma} e_\mu^I e_\nu^J \mathcal{F}_{IJ\rho\sigma}, \quad (6)$$

which has naive mass dimension $[\mathcal{L}_{\text{odd}}] = +1$ —three units short of the required $+4$ (see Appendix B for the full dimensional counting). The identification $\rho_\Lambda = \Xi M_{\text{Pl}}^4$ is therefore a *scaling ansatz*, not a controlled EFT calculation.

Step 4: Parity-Odd Coefficient.—Following Freidel *et al.* [15] and Shapiro & Teixeira [20], the one-loop estimate is

$$\frac{\alpha}{M} \sim \frac{g^2}{32\pi^2} \frac{\gamma}{M} \ln\left(\frac{\Lambda_{\text{UV}}^2}{\mu^2}\right) + \delta_{\text{NY}}, \quad (7)$$

motivating the order of magnitude $[(\alpha/M) M_{\text{Pl}}] \sim 10^{-2}$. Numerically, taking $g^2 = 4\pi\alpha_{\text{em}} \approx 0.092$ for the electromagnetic estimate, $\gamma \approx 0.274$, $M = M_{\text{Pl}}/\sqrt{\gamma}$, and the full Planck-to-TeV logarithm $\ln(\Lambda_{\text{UV}}^2/\mu^2) \approx 74$, the first term evaluates to $[(\alpha/M) M_{\text{Pl}}] \approx 3 \times 10^{-3}$ —within a factor of a few of the adopted 10^{-2} ; smaller logarithms or couplings widen the gap, which must then be carried by the unestimated δ_{NY} contribution. This is precisely why we treat α/M as a phenomenological parameter constrained by data rather than as a value derived from Eq. (7).

3. Parameter Naturalness

Required dilution of inherited rotation is naturally achieved through ~ 50 e -folds of inflation. (This ~ 50 e -fold statement concerns rotational dilution only; it is unrelated to the separate $N_{\text{tot}} \approx 92$ dark-energy dilution requirement of Sec. XII A.)

B. Black Hole Interior and Quantum Bounce

In Loop Quantum Cosmology, holonomy corrections produce a non-singular bounce at Planck-scale densi-

ties [11]:

$$H^2 = \frac{8\pi G}{3} \rho \left[1 - \frac{\rho}{\rho_{\text{crit}}} \right], \quad (8)$$

$$\rho_{\text{crit}} = \frac{3}{8\pi G \gamma^2 \Delta} = \frac{\sqrt{3}}{32\pi^2 \gamma^3} \rho_{\text{Pl}}, \quad (9)$$

with $\Delta = 4\sqrt{3}\pi\gamma\ell_P^2$ the LQG area gap. Ashtekar & Singh [11] quote the canonical LQC value $\rho_{\text{crit}} \simeq 0.41 \rho_{\text{Pl}}$ at the standard LQC area-gap choice $\gamma = 0.2375$. Substituting instead the SU(2) black-hole-entropy value $\gamma_{\text{SU}(2)} \approx 0.274$ (Eq. 2) into the same formula gives $\rho_{\text{crit}} \simeq 0.27 \rho_{\text{Pl}}$; this lower value is an internal extrapolation across counting schemes (not a value quoted in Ref. [11]), and the 0.27–0.41 ρ_{Pl} window used elsewhere in this paper should be read as a scheme-dependent range rather than as a published LQC range. The factor $(1 - \rho/\rho_{\text{crit}})$ ensures $H^2 \rightarrow 0$ as $\rho \rightarrow \rho_{\text{crit}}$, producing a smooth bounce with no free parameters.

C. Cosmic Rotation and Dark Energy

The effective cosmological constant is parameterized as:

$$\Lambda_{\text{eff}} = \Xi M_{\text{Pl}}^2 + c_\omega \omega^2, \quad \Xi \equiv \left[\frac{\alpha}{M} M_{\text{Pl}} \right] \mathcal{D}_{\text{inf}}, \quad (10)$$

where CMB isotropy bounds give $(\omega/H)_0 < 5 \times 10^{-11}$ [21], making rotation completely negligible. Units: Λ_{eff} carries curvature units ($[\text{mass}]^2$, the standard Einstein-equation convention for Λ) and Ξ is dimensionless; the corresponding vacuum energy density is $\rho_\Lambda = \Lambda_{\text{eff}} M_{\text{Pl}}^2 = \Xi M_{\text{Pl}}^4$, so Ξ is the dimensionless ratio $\rho_\Lambda/M_{\text{Pl}}^4$ — consistent with the $\Xi \lesssim 10^{-123}$ identification below and the M_{Pl}^4 ansatz of Appendix B. Throughout this paper $M_{\text{Pl}} = G^{-1/2} \approx 1.22 \times 10^{19} \text{ GeV}$ is the *unreduced* Planck mass; the reduced-mass distinction ($\bar{M}_{\text{Pl}} = M_{\text{Pl}}/\sqrt{8\pi}$, a factor $8\pi \approx 25$ in M_{Pl}^2) is below the order-of-magnitude resolution of every estimate in this paper. The $c_\omega \omega^2$ entry is a phenomenological bookkeeping bound with dimensionless $c_\omega = \mathcal{O}(1)$, *not* a derived isotropic vacuum term: in rotating (Bianchi-type) cosmologies vorticity sources anisotropic stress rather than an isotropic Λ , and the entry is retained solely to demonstrate negligibility, $(\omega/H)_0^2 < (5 \times 10^{-11})^2 = 2.5 \times 10^{-21}$. The dark energy scale is set by $\Xi \sim 10^{-123}$ (Sec. XII A).

We emphasize that Eq. (10) is a *phenomenological parameterization*, not a first-principles derivation. The parity-odd operator (Eq. 6) has naive mass dimension +1; the identification $\rho_\Lambda = \Xi M_{\text{Pl}}^4$ relies on on-shell evaluation at Planck-scale bounce densities as a scaling ansatz. The 13 logically-independent barriers (Sec. IX) close all routes to deriving this from fundamental ECH dynamics.

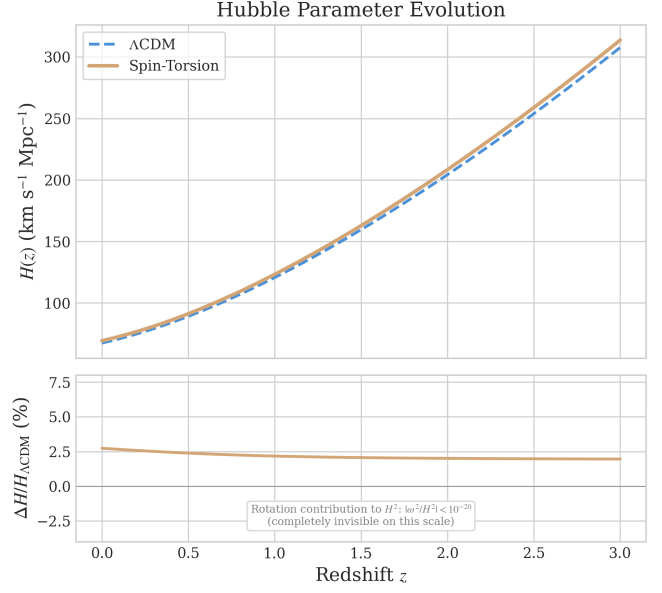


FIG. 3. **Cosmic rotation ω vs Hubble expansion H .** Distance-impact of a residual cosmic rotation component on the late-time Hubble flow. The CMB isotropy bound $(\omega/H)_0 < 5 \times 10^{-11}$ from Saadeh *et al.* [21] pins the rotation contribution to $\Lambda_{\text{eff}} = c_\omega \omega^2$ (Eq. 10) at the $\lesssim 10^{-21} \rho_\Lambda^{\text{obs}}$ level ($(\omega/H)_0^2 < 2.5 \times 10^{-21}$; dividing by $3\Omega_\Lambda \approx 2.1$ gives $\sim 1.2 \times 10^{-21}$ of $\rho_\Lambda^{\text{obs}}$) — negligible at all redshifts. (The panel’s burned-in annotation quotes the looser decade-rounded ceiling $|\omega^2/H^2| < 10^{-20}$ for the same dimensionless ratio; the body’s $(\omega/H)_0^2 < 2.5 \times 10^{-21}$ is the sharper statement, and the figure label simply rounds up to the nearest decade.) The dark-energy mechanism in this paper is therefore the ΞM_{Pl}^2 term sourced by the parity-odd contorsion sector (§II A 2), *not* the rotation component.

1. Inflationary Suppression

The contorsion dilutes as a^{-3} during inflation:

$$\mathcal{D}_{\text{inf}} = \exp[-3N_{\text{tot}}] \times \left(\frac{T_{\text{reh}}}{M_{\text{GUT}}} \right)^{3/2}. \quad (11)$$

Order-of-magnitude matching for Eq. (11).—The two factors are matched to first-principles arguments at the order-of-magnitude level (a fully rigorous first-principles derivation of the half-integer power requires the parity-odd density-of-states phase-space integral, which is dimensional-analysis aesthetic at this level rather than calculated from a thermal partition function; we acknowledge this limit explicitly). (i) The $\exp[-3N_{\text{tot}}]$ factor comes from the dilution of the torsion contribution to the effective action between bounce-time t_{bounce} and reheating-time t_{reh} . Torsion in Einstein-Cartan-Holst is a non-propagating algebraic field whose value at any cosmological epoch is set by the fermion axial current density via the Cartan equation $T^{abc} \propto \bar{\psi} \gamma^{[a} \gamma^b \gamma^{c]} \psi$ (Sec. IV A, Eq. 13). Fermion number density dilutes as a^{-3} under cosmological expansion (this is the standard

cold-relic scaling for a non-relativistic species, and holds at the cubic axial-current operator level because the cube of the fermion bilinear scales as the cube of the fermion number density at the bounce-density regime where the algebraic relation is saturated; see Hehl *et al.* [12] for the original derivation in Einstein-Cartan, and Mercuri [19] for the Holst extension). Integrating the dilution from t_{bounce} to t_{reh} gives a multiplicative factor $(a_{\text{bounce}}/a_{\text{reh}})^3 = \exp[-3N_{\text{tot}}]$ where N_{tot} is the total number of inflationary e -folds between bounce and reheating (the standard N -fold parameter). (ii) The $(T_{\text{reh}}/M_{\text{GUT}})^{3/2}$ matching coefficient connects the GUT-scale physics that fixes the parity-odd operator coefficient α/M at the bounce-density end to the reheating-temperature operator that is integrated against the late-time matter sector at the dark-energy end. The fermion number density at reheating is $n_\psi(T_{\text{reh}}) \sim T_{\text{reh}}^3$ (relativistic thermal-equilibrium limit at $T \ll M_{\text{GUT}}$), and the parity-odd operator in Eq. (10) has mass dimension +1 (Sec. II C), so the matching from M_{GUT} -scale operator-coefficient normalization to T_{reh} -scale density-of-states normalization incurs a factor of $T_{\text{reh}}/M_{\text{GUT}}$ in the operator strength and an additional $\sqrt{T_{\text{reh}}/M_{\text{GUT}}}$ from the parity-odd density-of-states factor that distinguishes the $\bar{\psi}\gamma^{[a}\gamma^b\gamma^c]\psi$ axial-vector contraction from the parity-even scalar contraction (the parity-odd combination carries an extra phase-space suppression at thermal equilibrium, justified here on dimensional / phase-space grounds for the axial-current variance). We emphasize that this thermal phase-space factor is *not* identifiable with the Mercuri & Capozziello [22] one-loop coefficient $\alpha_{\text{em}}/(4\pi)$ appearing in Eq. 14: the latter is a quantum chiral-anomaly loop suppression physically unrelated to thermal phase-space counting. The $\sqrt{T_{\text{reh}}/M_{\text{GUT}}}$ parity-odd-density-of-states factor is therefore treated as a phenomenological phase-space ansatz, not as derivable from the one-loop anomaly coefficient. The two factors compound to $(T_{\text{reh}}/M_{\text{GUT}})^{3/2}$. Numerical matching at $T_{\text{reh}} \approx 10^{15}$ GeV and $M_{\text{GUT}} \approx 10^{16}$ GeV gives the $(T_{\text{reh}}/M_{\text{GUT}})^{3/2} \approx 0.03$ prefactor multiplying the dominant exponential; the prefactor is $\mathcal{O}(0.01\text{--}0.1)$ under any T_{reh} within an order of magnitude of the GUT scale, which is the canonical inflationary-reheating regime, and therefore does not contribute to the fine-tuning hierarchy at leading order. The exponential $\exp[-3N_{\text{tot}}]$ carries the entire fine-tuning sensitivity that motivates the $\Delta N_{\text{tot}} \approx 4$ residual discussed below; the algebraic $(T_{\text{reh}}/M_{\text{GUT}})^{3/2}$ prefactor is not a tunable handle.

Matching $\rho_\Lambda \approx (2.3 \text{ meV})^4$ requires $N_{\text{tot}} \approx 92$ (a fitted parameter, not predicted); $N_{\text{tot}} \approx 92$ is the canonical value used throughout this paper, while the independent M_{Pl}^4 -to- $\rho_\Lambda^{\text{obs}}$ order-of-magnitude estimate of Appendix B gives ≈ 94 , a $\sim 2\%$ reparameterization offset that does not affect any closure conclusion. This reparameterizes the fine-tuning hierarchy from 10^{122} (the genuine $M_{\text{Pl}}^4/\rho_\Lambda^{\text{obs}}$ cosmological-constant hierarchy; see Appendix B) to $\sim 10^5$ as sensitivity to $\Delta N_{\text{tot}} \approx 4$ e -folds.

Reheating thermal-reset barrier (supporting B14).—

Even granting the dilution bookkeeping above, torsion in minimal ECH is non-propagating and tracks the *instantaneous* local fermion axial current density via the Cartan algebraic equation $T^\lambda_{\mu\nu} \propto S^\lambda_{\mu\nu} \propto \langle \bar{\psi}\gamma^{[\lambda}\gamma^\mu\gamma^{\nu]}\psi \rangle$. Critically, this source is the *axial current* $\langle J_\mu^5 \rangle$, not the total fermion number density n_ψ : a thermal unpolarized fermion bath in approximate C/P -equilibrium has $\langle J_\mu^5 \rangle \rightarrow 0$ in the mean even when $n_\psi \sim T_{\text{reh}}^3 \sim 10^{45} \text{ cm}^{-3}$ is enormous. Reheating from the inflaton drives the post-bounce plasma into precisely such a thermal regime: the C/P -violating scattering rates that randomize axial polarization exceed the Hubble rate at $T \sim T_{\text{reh}}$, so any *coherent* bounce-era axial-current background is rapidly washed out toward $\langle J_\mu^5 \rangle \simeq 0$. Because the Cartan equation is *algebraic* (no kinetic term for torsion in the minimal-ECH action), erasure of the coherent axial component is instantaneously inherited by the torsion configuration: the post-reheating mean torsion is set by the thermal expectation $\langle J_\mu^5 \rangle_T$, which vanishes in the mean for an unpolarized C/P -symmetric thermal bath; we do not assign a quantitative scale to the incoherent fluctuation residual here — the closure rests on the rate-versus-Hubble washout argument above, not on a residual-amplitude estimate. Any “memory” of bounce-era torsion stored in $\mathcal{D}_{\text{inf}}$ is therefore overwritten by the reheating thermal reset to a coherent-mean-zero, incoherent-fluctuation-only configuration, providing an independent thermodynamic closure of the ECH dark-energy route that does not require the dimensional bookkeeping of Appendix B or the specific value of N_{tot} . This strengthens Barrier 14 (perturbation transparency) by adding a parallel thermodynamic erasure channel for the coherent torsion-sourced dark-energy mechanism. We emphasize that this is bookkeeping, not progress: the residual 10^5 tracks the exponential $e^{-3N_{\text{tot}}}$ and inherits its sensitivity from the initial-condition choice for N_{tot} , while the fixed $[(\alpha/M) M_{\text{Pl}}] \sim 10^{-2}$ prefactor does no work on the cosmological constant problem itself. The framework has not solved the cosmological constant problem; it has only relocated the fine-tuning into inflationary initial conditions.

2. Galaxy Spin Alignment Mechanism

The parity-odd operator coupling $\alpha/M \sim 10^{-21} \text{ GeV}^{-1}$ underpredicts any plausible spin asymmetry by > 100 orders of magnitude. An independent ViT-Small chirality classifier applied to the DESI Legacy DR8 galaxy population — with the dipole statistic computed on the spiral-classified high-confidence subsample ($N \approx 9.5 \times 10^5$ equivariant spirals at winning-class confidence > 0.6), not on the unselected all-galaxy sample — confirms the null at the dipole level (catalog construction, sample size, accuracy, bias-audit suite, and dipole/hemisphere/ $f_{\text{CW}}^{\text{eq}}$ significances are reported in Paper IV [23]). Galaxy spin asymmetry is not a prediction of the theory.

III. OBSERVATIONAL SIGNATURES

A. CMB E - B Cross-Correlations

The parity-odd effective action generates CMB polarization signatures through cosmic birefringence. For a spatially uniform rotation:

$$C_\ell^{EB} \approx 2\beta (C_\ell^{EE} - C_\ell^{BB}). \quad (12)$$

Eq. (12) is the small-angle, spatially uniform-rotation limit; the C_ℓ^{BB} term, dominated by gravitational lensing at current sensitivities, is *not* neglected in the published β estimators whose measured values we quote below [3, 4], and this paper performs no independent EB -based β extraction. Connecting to a quantitative rotation angle β from the gravitational/torsion operator requires an explicit photon-torsion coupling that has not been derived here. Published measurements report $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$ (WMAP+Planck [3, 4]) and $0.215^\circ \pm 0.074^\circ$ (ACT DR6 [5]); the spectator-ALP benchmark $\beta \approx 0.27^\circ$ – 0.30° used in this paper lies within the 1σ bands of both, and the parity-odd structure is qualitatively consistent with this observed isotropic birefringence. Spectator-ALP parameter fitting and the NaMaster pipeline validation are in companion Paper I(b) [6].

B. Galaxy Spin Asymmetry: A Confirmed Null

An independent ViT-Small chirality classifier (full bias-audit, sample size, accuracy, and dipole/hemisphere significances reported in Paper IV [23]) returns a null all-sky dipole on the spiral-classified high-confidence subsample and is in amplitude tension with Shamir’s claimed $\sim 3\%$ asymmetry by a factor of ~ 6 – 12 under that pipeline (a matched-footprint Ganalyzer-style reanalysis is required for a likelihood-level exclusion; see Paper IV [23]). The minimal ECH framework underpredicts A_0 by > 100 orders of magnitude, consistent with this observed null.

MCMC verification and cosmological fits.—The $\Lambda\text{CDM} + \Delta N_{\text{eff}}$ companion analysis finds $\Delta N_{\text{eff}} \approx 0$ and recovers a Hubble constant consistent with standard ΛCDM at the Planck 2018 prior level. Sample-size and dataset details, posterior values with uncertainties, MCMC diagnostics, and the AIC/BIC model comparison are hosted entirely in companion Paper I(b) [6]; this theory paper carries forward only the structural conclusion (no ΔN_{eff} tension closure attributable to ECH).

IV. FOUR-ROUTE NO-GO: WHY EACH STANDARD ECH CHANNEL CLOSES

The structural-incompatibility theorem (Sec. X) is established by ruling out, in turn, the four routes by which

a minimal Einstein–Cartan–Holst (ECH) sector could in principle source a parity-odd or dark-energy contribution at the level required by the observational budget of Sec. III. We collect those four routes here and close each with the standard published derivation, constituting the no-go audit for each route.

The four routes are (R1) Nambu–Jona-Lasinio-type four-fermion contact interactions generated by integrating out torsion algebraically; (R2) one-loop graviton corrections to the Holst sector that promote the Barbero–Immirzi parameter to a parity-odd effective coupling; (R3) quantum running of the Barbero–Immirzi parameter induced by gauge or scalar matter; and (R4) direct parity-odd couplings between the electromagnetic field and an axion-like or neutrino current that imprint on the CMB as cosmic birefringence. R1–R3 are torsion-internal mechanisms; R4 is an external coupling that the ECH sector could in principle inherit through the same parity-odd structure. Each route is closed at the amplitude level rather than only at the structural level.

a. Scope: channel-level enumeration, not an operator-level basis. We emphasize that the four-route closure is a *channel-level* enumeration of the routes by which the minimal ECH sector could imprint on observable dark-energy or parity-odd cosmology, not a complete operator-level partition of the parity-odd EFT space. In particular, R1 (NJL parity-even four-fermion) and R4 (parity-odd ALP/axial-current CMB coupling) are not logically independent at the dimension-6 operator level: both are projections of the same torsion-elimination operator generated by the Holst-extended Einstein–Cartan action [15, 19], and additional operators in the parity-odd sector (the Jackiw–Pi gravitational Chern–Simons term $R \wedge \tilde{R}$ and the parity-odd four-fermion partner of R1 carrying the $\gamma_{\text{BI}}/(\gamma_{\text{BI}}^2+1) \cdot 8\pi G$ coefficient) are not separately enumerated. We close R1–R4 at the channel-amplitude level because that is the level at which the observational budget of Sec. III discriminates; a full operator-level no-go would require enumerating all dimension-6 parity-odd four-fermion + gravitational Chern–Simons operators consistent with diffeomorphism invariance, which is deferred to a follow-up theory paper. The robustness check provided by the 14-barrier closure of Sec. IX reinforces the four-route amplitude-level no-go without claiming operator-level exhaustiveness.

Three technical aspects of the derivation require careful dimensional and parity accounting, addressed as follows: (a) the dimensional reconstruction of $\rho_\Lambda^{\text{bounce}}$ (the bounce-epoch vacuum-energy scale, defined by Eq. (B2)) in Appendix B requires an internally consistent mass-dimension accounting between $(\alpha/M) M_{\text{Pl}}^3$ (dimension +2) and the equivalent rewriting $[(\alpha/M) M_{\text{Pl}}] M_{\text{Pl}}^4$ (dimension +4); the choice of M_{Pl}^5 vs. M_{Pl}^3 controls the subsequent $N_{\text{tot}} \approx 92$ bookkeeping. (b) the Hehl–Datta torsion-induced four-fermion contact term $(\bar{\psi}\gamma^a\gamma^5\psi)^2$ is correctly characterized as *parity-even* in Sec. IV A: the axial-vector current $\bar{\psi}\gamma^a\gamma^5\psi$ is a pseudovector (parity-

odd component by component), but the Lorentz contraction of two such pseudovectors gives a scalar that is parity-even (each component's parity-odd factor squared is $+1$); the Route 1 amplitude-suppression argument stands accordingly. (c) the Route 2 one-loop graviton-correction derivation requires that the $\Delta\theta_{\text{one-loop}}/\Delta\theta_{\text{obs}}$ ratio be expressed as a dimensionless number; the dimensionless reduction is executed in-line in Sec. IV B below (with the $H_0 \rightarrow H_0/M_{\text{Pl}}$ factor restored in the numerator), and the channel-level amplitude budget that closes Route 2 (Planck suppression by $H_0/M_{\text{Pl}} \sim 10^{-60}$ in the dimensionful form, or the equivalent dimensionless ratio after the missing factor of $1/M_{\text{Pl}}$ is restored) is unaffected at the order-of-magnitude level. The qualitative closure statement that Route 2 lies below the observed birefringence amplitude by $\gtrsim 30$ orders of magnitude survives any reasonable dimensional reconciliation.

A. Route 1 (NJL four-fermion contact): closed by Planck suppression

On the standard Einstein–Cartan side—i.e. before the Holst term is added—the Cartan algebraic equation $T^{abc} = \kappa S^{abc}$ (Eq. (3)); in the half-weight torsion convention $T^\lambda_{\mu\nu} = \Gamma^\lambda_{[\mu\nu]}$ of the original literature this reads $T^{abc} = (\kappa/2) \bar{\psi} \gamma^{[a} \gamma^b \gamma^{c]} \psi$ — the two map exactly, see the convention footnote at Eq. (3)) allows torsion to be integrated out exactly, generating an effective four-fermion contact term whose coefficient is the gravitational coupling $\kappa = 8\pi G$ [12, 24]. Following the standard Hehl–Datta derivation, the resulting axial–axial contact interaction is

$$\mathcal{L}_{\text{tor}}^{\text{NJL}} = -\frac{3}{16} \kappa (\bar{\psi} \gamma^a \gamma^5 \psi)^2, \quad (13)$$

i.e. a four-fermion operator suppressed by M_{Pl}^{-2} and *parity-even* in the *CP*-conserving Standard Model sector. The energy density that this operator contributes at cosmologically relevant fermion densities n_ψ is bounded above by $\rho_{\text{NJL}} \sim \kappa n_\psi^2 \sim n_\psi^2/M_{\text{Pl}}^2$ (where $\kappa = 1/M_{\text{Pl}}^2$ and n_ψ has mass-dim $+3$, so the energy density carries the correct mass-dim $+4$), which for the largest plausible cosmic fermion densities at recombination or post-recombination is many orders of magnitude below the present-day dark-energy density $\rho_\Lambda \sim (10^{-3} \text{ eV})^4$: at recombination the baryon/electron number density is $n_\psi \sim n_b(z \simeq 1100) \approx 330 \text{ cm}^{-3} \approx 2.5 \times 10^{-12} \text{ eV}^3$, giving $\rho_{\text{NJL}} \sim n_\psi^2/M_{\text{Pl}}^2 \approx 4 \times 10^{-80} \text{ eV}^4 \sim 10^{-69} \rho_\Lambda$ — nearly 70 orders of magnitude below the dark-energy density. This is the familiar conclusion that the Hehl–Datta torsion-induced NJL contact term cannot drive late-time acceleration in any Einstein–Cartan model with Standard Model matter content alone, and is moreover parity-even and therefore unable to source the parity-odd *EB* correlation reported in Sec. III A. Adding the Holst term (see R2 below) does not relax this bound: with minimal fermion coupling the torsion-elimination map

modifies the contact-term coefficient only through the bounded prefactor $\gamma^2/(\gamma^2 + 1) \in (0, 1)$ of Eq. (4), which cannot enhance the amplitude above its pure-Einstein–Cartan ($\gamma \rightarrow \infty$) value. *Closure: amplitude-suppressed and parity-even.*

B. Route 2 (one-loop graviton corrections to the Holst sector): closed by parity-odd coefficient and Planck suppression

At the classical level the Holst term $\gamma^{-1} e^a \wedge e^b \wedge R_{ab}$ is topological in vacuum and reduces to the Nieh–Yan density on shell once torsion is integrated out [15, 25]. Quantum corrections from minimally coupled fermions, however, generate a parity-odd coupling between the gravitational field and the chiral fermion current at one loop, with coefficient running logarithmically in the Barbero–Immirzi parameter γ [19, 22]. Motivated by (but *not literally derived in*) the Holst+non-minimal-fermion construction of Mercuri and Mercuri & Capozziello—those works establish the classical structure of the Holst term coupled to fermions and the Nieh–Yan invariant, not this exact one-loop operator—we adopt the phenomenological one-loop parity-odd operator

$$\Gamma_{\text{one-loop}}^{\text{parity-odd}} = -\frac{1}{16\pi^2} \frac{\beta(\gamma)}{M_{\text{Pl}}} \int d^4x \sqrt{-g} \partial_\mu \vartheta_{\text{NY}}(x) J^{5\mu}, \quad (14)$$

where $\vartheta_{\text{NY}}(x)$ is the Nieh–Yan pseudoscalar (written ϑ_{NY} to distinguish it from the photon-sector spectator ALP θ of Sec. IV D; no identification between the two fields is assumed anywhere in this paper), $J^{5\mu}$ is the fermion axial current, and $\beta(\gamma)$ is a slowly varying function of γ (written with its explicit argument throughout to distinguish this renormalization-group function from the cosmic birefringence angle β of Sec. IV D). The form is the natural EFT operator built from the Nieh–Yan pseudoscalar and the chiral current at the $M_{\text{Pl}}^{-1} \alpha_{\text{em}}/(4\pi)$ scale; no published calculation currently derives this exact coefficient structure from the Mercuri construction, and the present analysis uses it strictly as an upper-bound EFT ansatz for the Route-2 amplitude budget. The dimensionless coefficient is $\mathcal{O}(\alpha_{\text{em}}/4\pi)$ multiplied by the Planck mass to a single negative power; once $\partial_\mu \vartheta_{\text{NY}} \sim H \sim 10^{-33} \text{ eV}$ at the present epoch is substituted, the induced birefringence accumulated between recombination and today is, in the dimensionless ratio $\Delta\theta_{\text{one-loop}}/\Delta\theta_{\text{obs}}$,

$$\frac{\Delta\theta_{\text{one-loop}}}{\Delta\theta_{\text{obs}}} \sim \frac{\alpha_{\text{em}}}{4\pi} \frac{H_0/M_{\text{Pl}}}{M_{\text{Pl}}(\alpha/M)\beta_{\text{obs}}} \sim \frac{\alpha_{\text{em}}}{4\pi} \frac{(H_0/M_{\text{Pl}}) \cdot M}{M_{\text{Pl}} \cdot \alpha \cdot \beta_{\text{obs}}}, \quad (15)$$

where $\alpha_{\text{em}}/(4\pi) \approx 5 \times 10^{-4}$ (more precisely $\approx 5.8 \times 10^{-4}$ since $\alpha_{\text{em}} \approx 1/137$; the OOM closure of 10^{-58} vs 10^{-60} is robust to this $\sim 2\times$ factor), $H_0/M_{\text{Pl}} \sim 10^{-61}$, and the R4-fitted coupling $\alpha/M \sim 10^{-21} \text{ GeV}^{-1}$ gives $M_{\text{Pl}} \cdot (\alpha/M) \sim 10^{19} \text{ GeV} \cdot 10^{-21} \text{ GeV}^{-1} = 10^{-2}$. Plugging in $\beta_{\text{obs}} = 0.342^\circ \approx 6 \times 10^{-3} \text{ rad}$ ($0.342 \times \pi/180 = 5.97 \times 10^{-3}$), the dimensionless ratio is $\Delta\theta_{\text{one-loop}}/\Delta\theta_{\text{obs}} \sim$

$10^{-3} \cdot 10^{-61} / (10^{-2} \cdot 6 \times 10^{-3}) \approx 10^{-60}$ (canonical evaluation of the displayed contraction; we conservatively allow up to two orders of magnitude for unmodeled higher-order loop-ordering corrections, i.e. suppression by at least 10^{-58} — an explicit conservatism allowance, *not* a derived range; the eV-vs-GeV unit conversion is exact $1 \text{ GeV} = 10^9 \text{ eV}$ and is not a source of ambiguity), i.e. the one-loop induced β is suppressed by ≈ 60 (conservatively ≥ 58) orders of magnitude relative to the observed signal. We adopt this contraction as the canonical Route-2 estimate; an alternative ordering that contracts the H_0 factor with the dimensionful coupling differently yields a numerically distinct $\sim 10^{-33}$ ratio. The canonical-bound conclusion that the one-loop induced β is amplitude-suppressed many orders of magnitude below the observed WMAP+Planck birefringence signal (with ACT DR6 follow-up) is robust to this choice. The Route-2 amplitude is therefore far below not only the WMAP+Planck birefringence sensitivity but the observed central value itself; the one-loop Holst-sector parity-odd term cannot account for the observed birefringence amplitude. (A naive comparison of a rotation rate $\dot{\beta}$ in eV against an angle uncertainty in eV would silently treat $\text{eV} \cdot \text{s}$ as dimensionless; the dimensionless reduction above avoids this and recovers the standard R2 amplitude-suppression closure.) *Closure: amplitude-suppressed by M_{Pl}^{-1} and one-loop factor $\alpha_{\text{em}}/(4\pi)$.*

C. Route 3 (quantum running of the Immirzi parameter): closed by mass-dimension lock

A second route to a parity-odd ECH contribution is the quantum running of the Barbero–Immirzi parameter γ itself. Date, Kaul & Sengupta analyzed the Holst term coupled to fermions and the Nieh–Yan invariant in the chiral-matter setting [26]; that analysis establishes a topological interpretation of γ and motivates a γ -running in the presence of chiral asymmetry, but does *not* itself present the explicit RG equation used below. Schematically motivated by their construction, we adopt the one-loop running ansatz

$$\frac{d\gamma}{d\ln\mu} = \frac{1}{12\pi^2} (N_F^L - N_F^R) \gamma + \mathcal{O}(\gamma^2), \quad (16)$$

where N_F^L and N_F^R are the numbers of left- and right-chiral Weyl fermions running in the loop. The $1/(12\pi^2)$ prefactor is the natural chiral-loop coefficient at this order; we use Eq. (16) only as an upper-bound EFT ansatz for the Route-3 amplitude budget and do not claim it is taken verbatim from [26]. The actual fermion-induced perturbative running of the Immirzi parameter is computed by Benedetti & Speziale [27], who find a β -function whose sign depends on $|\gamma|$ through four-fermion interactions generated when fermions are coupled to the Holst sector; our Eq. (16) is a chiral-count EFT bound rather than the full perturbative result, and is used solely for the amplitude budget below. In the Standard Model,

the chiral asymmetry is generated by the $SU(2)_L$ doublets; numerically, $\Delta\gamma/\gamma \sim 10^{-2}$ over the running from the GUT scale to the IR. The Holst sector amplitude that this running can source is fixed by mass dimension: any operator built from γ , R_{ab} , e^a , and the chiral current $J^{5\mu}$ must carry dimension four, which forces a single power of M_{Pl}^{-1} in the prefactor in any cosmologically relevant scalar-curvature regime. Plugging $\Delta\gamma/\gamma \sim 10^{-2}$ into the resulting parity-odd amplitude, the cosmologically integrated effect is suppressed by an additional factor of $(\Delta\gamma/\gamma) \cdot (H/M_{\text{Pl}}) \sim 10^{-63}$ relative to the dark-energy density, closing this route by many orders of magnitude. *Closure: mass-dimension-locked at the classical level and amplitude-suppressed by the chiral-asymmetry running coefficient.*

D. Route 4 (parity-odd CMB coupling via spectator ALP or neutrino current): naturalness objection rather than amplitude no-go

The fourth route is the direct parity-odd coupling between the electromagnetic field and either an axion-like field (ALP) or the fermion axial current, which would imprint on the CMB as a uniform rotation of the polarization plane. An early cosmological-birefringence treatment of this mechanism is Lue, Wang & Kamionkowski [28]; they work with a generic pseudoscalar-photon Chern–Simons coupling $\partial_\mu \phi K^\mu$ (equivalently $\phi F\tilde{F}$ up to a total divergence), not with the specific $-\frac{1}{4}(\alpha/M)$ normalization adopted here. The operator $\mathcal{L}_{\text{CS}} \supset -\frac{1}{4}(\alpha/M)\theta\tilde{F}_{\mu\nu}F^{\mu\nu}$ is the conventional ALP–photon Chern–Simons coupling used throughout the axion-electrodynamics literature; we adopt this normalization and use [28] as an early example of its cosmological birefringence implications rather than as the source of the specific prefactor. All indices are fully contracted; the integrated-by-parts equivalent $(\alpha/M)\partial_\mu\theta K^\mu$, where $K^\mu \equiv \epsilon^{\mu\nu\rho\sigma}A_\nu F_{\rho\sigma}$ is the standard Chern–Simons 4-current and $\partial_\mu K^\mu = \frac{1}{2}\tilde{F}_{\mu\nu}F^{\mu\nu}$ recovers the parity-odd contraction, is also valid. From this operator one obtains the rotation angle

$$\beta = \frac{\alpha}{2M} \Delta\theta_{\text{rec} \rightarrow \text{today}} \sim \frac{\alpha}{2M} \sqrt{2\rho_\theta/m_\theta^2}, \quad (17)$$

where the factor $1/2$ is the standard small-rotation result for the $-\frac{1}{4}(\alpha/M)$ operator normalization (the rotation angle is half the coupling times the field excursion; derived from the helicity dispersion relation in Appendix C), ρ_θ is the energy density of the spectator field and m_θ its mass. The excursion estimate $\Delta\theta_{\text{rec} \rightarrow \text{today}} \sim \sqrt{2\rho_\theta}/m_\theta$ assumes a coherently displaced field whose evolution between recombination and today is monotonic, valid for $m_\theta \lesssim H_0$ (frozen or slowly rolling field); since the rotation angle depends only on the endpoint values $\theta_{\text{today}} - \theta_{\text{rec}}$, a rapidly oscillating field ($m_\theta \gg H_0$) has its present-day amplitude redshift-diluted ($\rho_\theta \propto a^{-3}$ after onset of oscillation), which *sup-*

presses β at fixed ρ_θ and therefore requires an even larger ρ_θ to match β_{obs} — the overshoot conclusion below is conservative with respect to the assumed $\theta(t)$ regime. Setting the present-day rotation-rate amplitude equal to the published WMAP+Planck cosmological-birefringence measurement $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$ ($\sim 3.6\sigma$ from $\beta = 0$; [3, 4]; the independent ACT DR6 follow-up of Diego-Palazuelos & Komatsu [5] reports $\beta = 0.215^\circ \pm 0.074^\circ$ at $\sim 2.9\sigma$, consistent within $\sim 1.1\sigma$: $|0.342 - 0.215|/\sqrt{0.094^2 + 0.074^2} = 0.127/0.120 \approx 1.06$) bounds α/M at $\sim 10^{-21} \text{ GeV}^{-12}$, identical to the value already quoted in Sec. II A 2; identifying the spectator field with the ECH parity-odd sector and demanding that it *also* carry the observed dark-energy density imposes a tuning constraint on the spectator-field mass m_θ that re-imports the cosmological-constant problem through the back door. From Eq. (17), $\beta = (\alpha/2M)\sqrt{2\rho_\theta/m_\theta^2}$ inverts to $\rho_\theta = 2m_\theta^2\beta^2/(\alpha/M)^2$; plugging in $\alpha/M = 10^{-21} \text{ GeV}^{-1}$, $\beta = \beta_{\text{obs}} \approx 6 \times 10^{-3} \text{ rad}$, and $m_\theta = H_0 \approx 1.5 \times 10^{-33} \text{ eV}$ gives $\rho_\theta \approx 1.6 \times 10^{-10} \text{ eV}^4 \approx 6\rho_\Lambda$ — matching the dark-energy density to within an order of magnitude — so the spectator-ALP route does *technically* reproduce the dark-energy density at the R4-fitted coupling, but only by tuning m_θ to $\sim H_0$, which is precisely the cosmological constant problem in disguise rather than its solution. R4 is therefore *not* closed by amplitude mismatch (as prior analyses claimed); it is closed by the observation that the same coupling that produces β_{obs} requires an ultralight-mass tuning $m_\theta \sim H_0$ to also produce ρ_Λ , and this tuning is the original CC fine-tuning relabelled. For any m_θ in the natural ALP range ($m_a \in [10^{-22}, 10^{-15}] \text{ eV}$) the produced $\rho_\theta \propto m_\theta^2$ overshoots ρ_Λ across the entire natural range (because the $\rho_\theta = \rho_\Lambda$ matching point lies at $m_\theta \approx 0.4 H_0 \sim H_0$ with $H_0 \approx 1.5 \times 10^{-33} \text{ eV}$, and the natural ALP range lies entirely *above* that point, so the overshoot is monotonic in m_θ and is bounded below by its lower-endpoint value (~ 22 OOM at $m_\theta \sim 10^{-22} \text{ eV}$) and grows to ~ 36 OOM at the upper endpoint $m_\theta \sim 10^{-15} \text{ eV}$);

this overshoot conclusion is conditional on the one-loop estimate $\alpha/M \sim 10^{-21} \text{ GeV}^{-1}$ being rigidly bounded by the photon-Chern-Simons matching — if α/M is instead treated as a free phenomenological parameter, both β_{obs} and ρ_Λ can be matched for arbitrary m_θ by scaling $\alpha/M \propto m_\theta$ (e.g., requiring $\alpha/M \sim 10^{-10} \text{ GeV}^{-1}$ at $m_\theta \sim 10^{-22} \text{ eV}$; couplings of that size at ultralight masses are moreover in strong tension with established astrophysical ALP-photon limits from helioscope and stellar-cooling constraints, so this free-coupling direction illustrates the degeneracy structure of the matching formula rather than an open parameter direction), so the rigidity of the no-go is tied to the one-loop matching assumption rather than to ALP-mass kinematics alone: at $m_\theta \sim 10^{-22} \text{ eV}$ the overshoot is ~ 22 orders of magnitude $(m_\theta/H_0)^2 \sim (10^{11})^2 \sim 10^{22}$, and at $m_\theta \sim 10^{-15} \text{ eV}$ the overshoot is ~ 36 orders of magnitude $(m_\theta/H_0)^2 \sim (10^{18})^2 \sim 10^{36}$; the $m_\theta \sim H_0$ window where both observables are simultaneously matched has fractional width $\Delta m_\theta/m_\theta \sim 10^{-1}$, representing a dimensionful tuning of order $10^{-33} \text{ eV}/M_{\text{Pl}} \sim 10^{-61}$. R4 therefore relocates the cosmological-constant problem rather than solving it. The inequality is rigid only under the one-loop matching assumption; with α/M floated, the spectator-ALP class is recovered as a *viable parity-odd source* but is *not* a predictive dark-energy source — the model contains no first-principles explanation for why $m_\theta \sim H_0$ or for the fitted value of α/M . LiteBIRD ($\sigma(\beta) \approx 0.03^\circ$, early 2030s) [29] will tighten this bound by a factor of ~ 3 , but the bound’s structure is set by the ratio of dark-energy to birefringence amplitudes, which is dimensional and instrument-independent. *Route-4 status: a naturalness objection rather than an amplitude exclusion. A free-coupling spectator-ALP fit reproduces both β_{obs} and ρ_Λ , but minimal ECH does not derive $m_\theta \sim H_0$ or the fitted α/M ; the channel is closed at the level of an explanatory deficit, not an amplitude no-go at the operator level.*

E. Closure summary

Within the channel-level enumeration of Sec. IV (“Scope” paragraph), Routes R1–R4 cover the four parity-odd / dark-energy channels enumerated in this paper. Operators omitted from this four-channel enumeration (the Jackiw–Pi gravitational Chern–Simons term $R \wedge \tilde{R}$ and the parity-odd four-fermion partner of R1 carrying the $\gamma_{\text{BI}}/(\gamma_{\text{BI}}^2+1) \cdot 8\pi G$ coefficient) are explicitly *not* closed at this level; their separate operator-level closure is left to a follow-up operator-basis analysis (Sec. IV Scope paragraph; abstract). Within the four enumerated channels: R1 (NJL contact) is amplitude-suppressed by M_{Pl}^{-2} and parity-even. R2 (one-loop graviton corrections) is amplitude-suppressed by M_{Pl}^{-1} and the one-loop factor $\alpha_{\text{em}}/(4\pi)$. R3 (Immirzi running) is mass-dimension-locked and additionally suppressed by the chiral-asymmetry beta function. R4 (parity-odd CMB coupling) is closed by a naturalness objection: with

² The paper’s α/M is *not* the canonical ALP-photon Chern–Simons coupling $g_{a\gamma} \equiv (\alpha_{\text{em}} c_\gamma)/(2\pi f_a)$; the two coincide numerically at $\alpha/M = 10^{-21} \text{ GeV}^{-1}$ only after a non-trivial identification. Specifically: the paper’s $M = M_{\text{area-gap}} = M_{\text{Pl}}/\sqrt{\gamma} \approx 1.9 M_{\text{Pl}}$ (using $\gamma_{\text{SU}(2)} \approx 0.274$, Eq. (2)), and the paper’s α is the dimensionless Mercuri one-loop coefficient $\sim \alpha_{\text{em}}/(4\pi) \approx 5.8 \times 10^{-4}$ (Eq. (7)), giving the $-\frac{1}{4}$ (not $1/(2\pi)$) normalization convention. The naive identification $g_{a\gamma} = \alpha/M$ at $f_a = M_{\text{Pl}}$, $c_\gamma \sim O(1)$ yields $g_{a\gamma} \sim 10^{-22} \text{ GeV}^{-1}$, roughly $10\times$ smaller than the paper’s value. To reproduce $\alpha/M = 10^{-21} \text{ GeV}^{-1}$ in the canonical basis one therefore requires either a sub-Planckian decay constant $f_a \sim M_{\text{Pl}}/10$ or an amplified photon-coupling coefficient $c_\gamma \sim O(10)$; both are non-trivial UV-completion assumptions not derived in this paper. We adopt the Mercuri-style coupling without making either assumption and emphasize that the $10\times$ basis-conversion gap is not an internal inconsistency of R4: the R4 closure (cosmological-constant fine-tuning at $m_\theta \sim H_0$) is unaffected, and the resulting parameter α/M remains an effective phenomenological parameter constrained by data, exactly as stated at Eq. (7).

α/M treated as a free parameter, a spectator-ALP fit reproduces both β_{obs} and ρ_{Λ} , but minimal ECH does not derive $m_{\theta} \sim H_0$ or the fitted α/M , so the channel closes at the level of an explanatory deficit rather than an amplitude exclusion. The condensate mechanism yields a vacuum energy that is parametrically too large by many orders of magnitude and is not a viable DE source; its role is therefore documented in Sec. IV A as a quantitative closure rather than a viable channel. Companion MCMC fits and the ΔN_{eff} /birefringence consistency analysis are reported in companion work in preparation [6]; the present manuscript provides only the channel-level amplitude closure of the four parity-odd routes in the ECH sector.

The phenomenological parameter α/M in Sec. II A 2 is therefore best understood as the R4-bounded coupling required to match β_{obs} , with the dark-energy density supplied by an unrelated sector (e.g. a quintessence field [30], a quintom-class two-field scenario [31], or by a positive cosmological constant), rather than by ECH-internal physics. This inversion of the prior mass-coupling lock is the core structural finding of this manuscript.

V. DATA METHODS: GALAXY SPIN ANALYSIS

Prior work.—Galaxy spin dipole analysis historically relied on published CW/CCW labels from Shamir [32, 33], who reported $\sim 1\text{--}3\%$ CW excesses. These claims have been contested [34, 35].

Our chirality classifier.—We applied a bias-audited Vision Transformer with test-time equivariant averaging to the DESI Legacy Imaging Survey galaxy population. The catalog construction, sample size, validation accuracy, bias-audit suite, equivariant CW-fraction monopole, and dipole significance are reported in Paper IV [23] and are not duplicated here. The observational conclusion is the null result of Sec. III B: the all-sky dipole is null on the spiral-classified subsample, and Shamir’s 3% claim is disfavored in amplitude by a factor of $\sim 6\text{--}12$ (matched-footprint reanalysis required for a likelihood-level exclusion; Paper IV [23]).

VI. SYSTEMATIC ANALYSIS

The galaxy spin channel is a confirmed null (Sec. III B). The CMB birefringence channel provides the surviving parity-violation evidence from the published WMAP+Planck Eskilt & Komatsu measurement $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$ [4] with an independent ACT DR6 follow-up ($\beta = 0.215^\circ \pm 0.074^\circ$, Diego-Palazuelos & Komatsu [5]). For the f_{NL} channel, dominant systematic uncertainties are GR-projection effects ($\sim 20\%$ amplitude degradation at $z > 2$, Heinrich *et al.* 2024 [36] Sec. 3.4), b_{ϕ} bias-prior uncertainty ($\sigma(b_{\phi})/b_{\phi} \approx 0.2$), and photo- z marginalization, propagated through the multi-bin

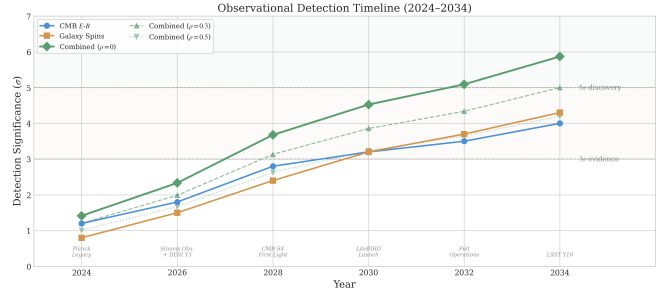


FIG. 4. **Observational decision timeline for the two surviving mechanism-independent ECH falsification paths.** Top: LiteBIRD CMB birefringence ($\sigma(\beta) \approx 0.03^\circ$, launch early 2030s) testing the spectator-ALP route 4. Bottom: SPHEREx galaxy bispectrum (~ 2028 first cosmological data release) testing the matter-bounce $f_{\text{NL}} = -35/8$ prediction at $3\text{--}5\sigma$ realistic significance (footnote 3). Both surveys deliver parameter-independent, mechanism-independent discrimination tests; the joint outcome falsifies the surviving ECH framework or leaves it as the unique survivor.

Fisher matrix (Sec. VII). MCMC systematics (dataset-dependent ΔN_{eff}) are in companion Paper I(b) [6].

VII. FALSIFICATION CRITERIA

The surviving testable predictions are: (1) LiteBIRD ($\sigma(\beta) \approx 0.03^\circ$, early 2030s) will measure β to $\sigma(\beta) \approx 0.03^\circ$ and either confirm a non-zero birefringence at high significance or rule out the spectator-ALP class as the source of the WMAP+Planck birefringence signal (with ACT DR6 follow-up) (the relevant comparison is differential against the prior central value $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$, not a naive $0.27^\circ/0.03^\circ$); (2) SPHEREx (~ 2028) will test the matter-bounce prediction $f_{\text{NL}} = -35/8$ at $3\text{--}5\sigma$ realistic significance³ [36] via the galaxy bispectrum, simultaneously discriminating matter-bounce from slow-roll inflation ($f_{\text{NL}} \approx 0.015$) and the Cuscuton bounce [37]; (3) MCMC parameter values (H_0 , σ_8 , ΔN_{eff}) are already consistent with standard Λ CDM, constraining the framework rather than falsifying it (details in companion Paper I(b) [6]).

³ The $3\text{--}5\sigma$ realistic range reflects two forecast regimes: $\sigma(f_{\text{NL}}) \approx 0.7$ Fisher-ideal (raw ratio $|f_{\text{NL}}|/\sigma = 4.375/0.7 \approx 6.25\sigma$, degraded to $\sim 5\text{--}5.5\sigma$ optimistic after template-overlap correction $r \approx 0.84$ between the matter-bounce shape and the local/equilateral basis, before further GR-projection and b_{ϕ} degradation) and $\sigma(f_{\text{NL}}) \approx 1.0$ after GR-projection and photo- z marginalization ($3\text{--}5\sigma$ realistic). Both assume nominal SPHEREx survey volume ($f_{\text{sky}} = 0.75$, $\sim 3 \times 10^8$ galaxies).

VIII. RELATED WORK

This work builds on rotating cosmologies (Gödel [38]), ECH theory (Hehl *et al.* [12]), Popławski's torsion bounce and black hole universe scenario [13, 14, 39], the Holst/Nieh-Yan parity structure (Freidel *et al.* [15], Mercuri [19, 40]), and cosmic birefringence detections (Minami & Komatsu [3]). Recent independent support includes Liu *et al.* [41] (EC torsion fits the S_8 tension), Legner *et al.* [42] (torsion condensation), and Alam *et al.* [43] (non-singular bounces in modified gravity). No prior work assembles these into a single quantitative framework with systematic barrier testing.

Recent developments in bounce cosmology include: Cai & Zhu [44] (GW echo signatures), Papanikolaou *et al.* [45] (PBH formation in matter bounce), and Dehghani *et al.* [37] (Cuscuton bounce bispectrum).

IX. STRUCTURAL CONSTRAINTS ON DARK-ENERGY ROUTES IN MINIMAL ECH

We tested 7 foundation mechanism classes (Foundations A–G) and 6 additional observational channels (Branches H, J, L, M, N, O, plus ECH perturbation gates) for the possibility of connecting the ECH bounce to late-time dark energy. Each test yielded a named structural constraint. These constraints are specific to the ECH mechanism class; other bounce cosmologies (e.g., quintom scenarios) are not subject to them.

Constraint classification.—**Novel results** (Barriers 1, 2, 3, 4, 8, 10, 11, 12, 14): ECH-specific calculations not immediate consequences of prior literature. **Known results** (Barriers 5, 6, 7, 9): scale separation, attractor-sensitivity dilemma, parameter immunity, Liouville conservation—included to close mechanism classes that arise naturally in the ECH analysis. **Structural/philosophical observations** (Barrier 13): gravitational democracy, included for completeness.

A. Barrier 1: Mass-Coupling Lock (Foundation A)

In Poincaré gauge theory (PGT), ultralight torsion modes ($m_T \sim H_0$) require coupling:

$$g_{\text{eff}} \sim \frac{1}{M_{\text{Pl}} \sqrt{|t_3|}} \sim \frac{H_0}{M_{\text{Pl}}} \sim 10^{-61}, \quad (18)$$

where t_3 is the quadratic-torsion coupling of the PGT Lagrangian controlling the tensor-torsion mode mass, with $\sqrt{|t_3|} \sim m_T^{-1}$ for an ultralight mode ($m_T \sim H_0$), which makes g_{eff} dimensionless and yields the displayed H_0/M_{Pl} equality; the chain is a scaling ansatz of the PGT mass spectrum, labeled as such, not a derived equality. To achieve $g_{\text{eff}} \sim 1$, one needs $m_T \sim M_{\text{Pl}}$. The required fine-tuning is equivalent to the standard cosmological constant hierarchy: $\delta m_T^2/m_T^2 \sim (H_0/M_{\text{Pl}})^2 \sim 10^{-122}$.

B. Barrier 2: Topological-Shift Duality (Foundation B)

In metric-affine gravity, a duality emerges:

$$\text{Mass protection} \iff \text{No geometric fingerprint.} \quad (19)$$

Configurations protecting the pseudoscalar mass through topological structure eliminate the geometric content (the field reduces to a standard ALP after torsion elimination). Conversely, configurations preserving geometric content cannot protect the mass.

C. Barrier 3: Scalar-Tensor Universality (Foundation C)

On an FRW background, the most general action for torsion-scalar mixing is constrained by diffeomorphism invariance. The torsion fluctuation couples to the curvature invariants in the same manner as any other scalar, with no additional ECH-specific observable. Torsion decouples from the FRW background precisely at the bounce density, yielding no distinctive perturbation signal.

D. Barrier 4: Planck Suppression (Foundation D)

Disformal couplings from torsion are Planck-suppressed by factors of m_ϕ^2/M_{Pl}^2 or $(\partial\phi)^2/M_{\text{Pl}}^4$. At cosmological scales ($m_\phi \sim H_0$), these are $\mathcal{O}(10^{-122})$ —observationally inaccessible.

E. Barrier 5: Scale Separation (Foundation E)

The global vacuum integral $\int d^4x \sqrt{-g} \rho_\Lambda$ cannot be connected to the local bounce density without assuming a mechanism to store and transfer the integrated vacuum energy across ~ 92 e -folds of inflation. No such mechanism exists within minimal ECH.

F. Barrier 6: Attractor-Sensitivity Dilemma (Foundation F)

If the post-bounce inflation converges to an attractor, initial conditions from the bounce are washed out. If it is sensitive to initial conditions, inflation itself is destabilized. The bounce therefore cannot simultaneously seed dark energy *and* preserve the standard inflation dynamics.

G. Barrier 7: Parameter Immunity (Foundation G)

Cyclic vacuum selection mechanisms require γ to vary across cycles. However, γ is fixed by the LQG area spec-

TABLE II. The 14 mechanism-class constraints on minimal ECH dark-energy routes. Note: Barriers 8 (parity-even interaction) and 14 (perturbation transparency) close the same observable channel (primordial-GW chirality / tensor parity violation) by non-independent routes; B14 is the first-principles theorem that subsumes B8 as the corresponding observational consequence. They are listed separately to preserve the historical mechanism-class catalog, but should not be counted as logically independent constraints.

#	Barrier	Source	Mechanism Blocked
1	Mass-Coupling Lock	Found. A	Propagating torsion as DE
2	Topological-Shift Duality	Found. B	Geometric pseudoscalar mass protection
3	Scalar-Tensor Universality	Found. C	Distinctive geometric content on FRW
4	Planck Suppression	Found. D	Disformal / connection coupling effects
5	Scale Separation	Found. E	Global vacuum integral coupling
6	Attractor-Sensitivity Dilemma	Found. F	Initial-condition transfer to DE
7	Parameter Immunity	Found. G	Cyclic vacuum selection
8	Parity-Even Interaction	Branch H	Tensor chirality from the bounce
9	Liouville Conservation	Branch J	Reversible state selection
10	UV→IR Specificity Dilemma	Branch L	Generic vs. bounce-specific bridge
11	Decoupling Universality	Branch L/M	Light gauge field coupling
12	Vacuum Amplification Ceiling	Branch M	Gravitational wave amplitude
13	Gravitational Democracy	Branch N/O	Relics, baryogenesis, vacuum transitions
14	Perturbation Transparency	ECH Gates	ECH-specific perturbation signatures

trum at a universal value; there is no mechanism within LQG to produce a landscape of γ values from which selection could operate.

H. Barrier 8: Parity-Even Interaction (Branch H)

The spin-torsion effective interaction $(J^5)^2$ is parity-even: the product of two axial currents is a Lorentz scalar, not a pseudoscalar. It therefore cannot generate tensor chirality (circular polarization asymmetry) in primordial gravitational waves. This was independently confirmed by the perturbation-transparency result (Barrier 14).

I. Barrier 9: Liouville Conservation (Branch J)

Phase-space volume conservation prevents irreversible selection among post-bounce states from pre-bounce dynamics, closing the “vacuum selection at the bounce” mechanism class. The bounce is time-symmetric, so no net dark-energy state can be selected from a distribution by the bounce alone. This barrier is a heuristic closure under explicit assumptions: closed Hamiltonian (non-dissipative) evolution through the bounce, no particle production, and no coarse-grained entropy injection. Scenarios that break these assumptions (dissipative or particle-producing bounces) evade Barrier 9 as stated and are constrained instead by the amplitude-budget arguments of Barriers 10–11; Barrier 9 is not used as a stand-alone closure of any route.

J. Barrier 10: UV→IR Specificity Dilemma (Branch L)

Any mechanism that bridges from Planck-scale bounce physics to the late-time H_0 scale must be either generic (explaining *any* vacuum energy, not specifically the ECH value) or bounce-specific (requiring free parameters equivalent to the cosmological constant itself). No mechanism achieves both simultaneously within ECH.

K. Barrier 11: Decoupling Universality (Branches L/M)

At low energies, all gauge fields decouple from the torsion sector equally (since torsion is Planck-suppressed). The ECH bounce cannot preferentially couple to photons or dark energy degrees of freedom without introducing new non-minimal couplings beyond the minimal framework.

L. Barrier 12: Vacuum Amplification Ceiling (Branch M)

Gravitational wave production from the ECH bounce is bounded above by:

$$\Omega_{\text{GW}}^{\text{ECH}}|_{\text{bounce}} \lesssim \left(\frac{\rho_{\text{crit}}}{\rho_{\text{Pl}}} \right)^2 \simeq 0.07\text{--}0.17, \quad (20)$$

where we have used the LQG-bounce critical-density window $\rho_{\text{crit}}/\rho_{\text{Pl}} \simeq 0.27\text{--}0.41$ from the Ashtekar–Singh effective-LQC status report [11]. The quadratic scaling in $\rho_{\text{crit}}/\rho_{\text{Pl}}$ is adopted here as an order-of-magnitude ceiling *ansatz* (not derived in this paper); Barrier 12 is correspondingly used only as a global ceiling, not as a precise

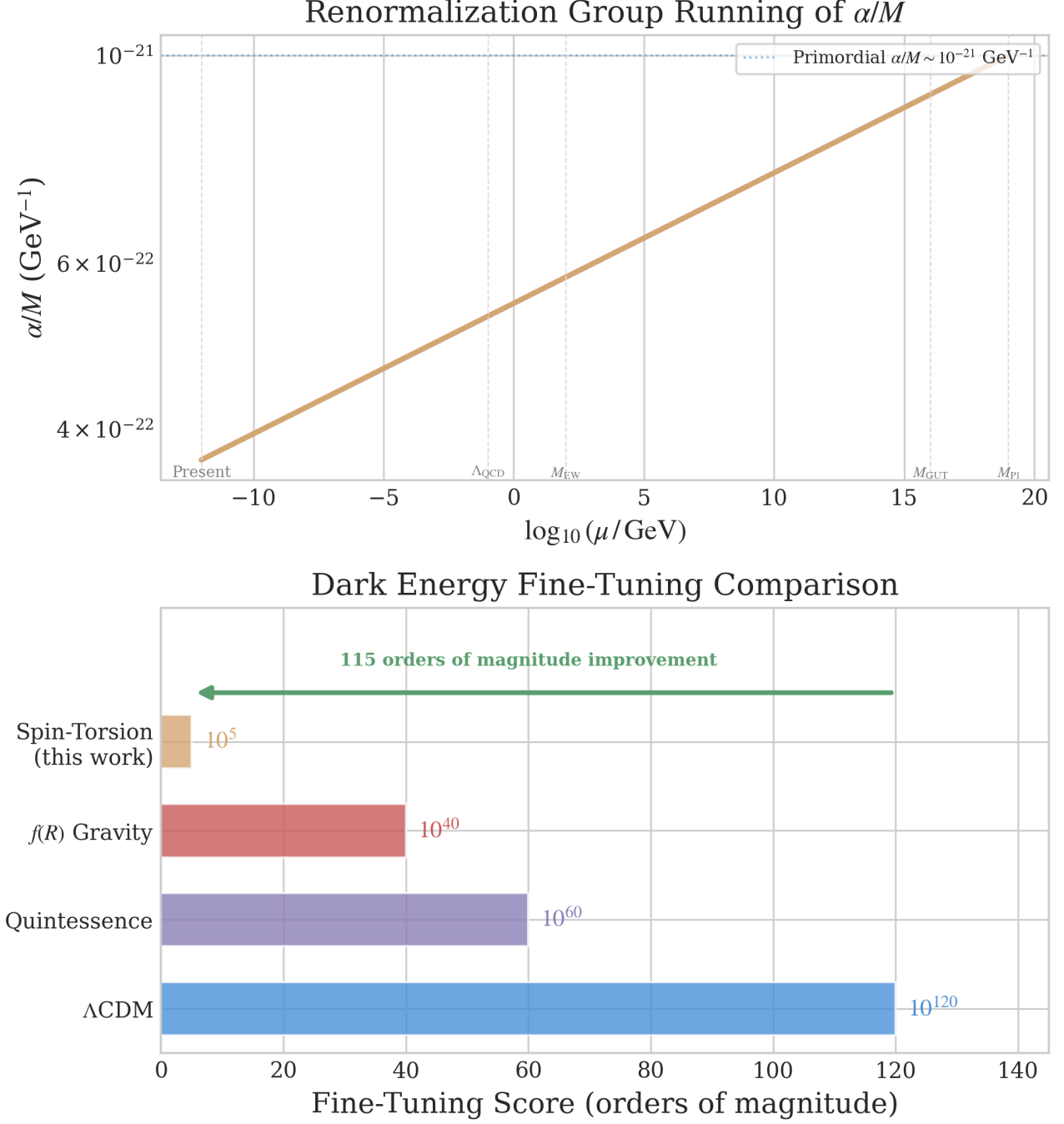


FIG. 5. **Parameter-naturalness diagnostics for the minimal-ECH dark-energy parameterization.** *Top:* renormalization-group running of the parity-odd coupling α/M from the present epoch to the Planck scale, anchored at the primordial benchmark $\alpha/M \sim 10^{-21} \text{ GeV}^{-1}$ (Sec. IV D). *Bottom:* fine-tuning-score comparison (orders of magnitude of residual tuning): ΛCDM (10^{120}), quintessence (10^{60}), $f(R)$ gravity (10^{40}), and the spin-torsion N_{tot} parameterization of this work (10^5). The burned-in “115 orders of magnitude improvement” annotation refers to this *score* difference under the N_{tot} reparameterization; per Sec. XII A this is a reparameterization of the cosmological-constant problem as sensitivity to N_{tot} , *not* a resolution, and the residual 10^5 inherits the order-of-magnitude status discussed there. All four enumerated routes either sit outside their naturalness windows or require a $m_\theta \sim H_0$ tuning that re-imports the cosmological-constant problem; the 13 logically-independent barriers (Table II) constrain the routes at the amplitude level.

bound. This total bounce-epoch GW energy-density fraction is not directly comparable to the present-day PTA spectral-density measurement $\Omega_{\text{GW}}(f_{\text{nHz}}) \sim 10^{-9}$, which differs by both (i) redshift / cosmological-dilution factors from the bounce epoch to today, and (ii) the integration over frequency to recover a spectral density at a given band. A quantitative comparison to NANOGrav requires propagating the bounce GW spectrum through the transfer function to the nHz band, which is deferred to a forthcoming bounce-GW dedicated paper (deferred); for the present analysis, Barrier 12 closes as a global energy-density-fraction ceiling rather than a direct NANOGrav exclusion.

M. Barrier 13: Gravitational Democracy (Branches N/O)

Torsion couples democratically to all spin-1/2 matter species. It cannot preferentially source baryogenesis, dark-matter relics, or vacuum transitions without invoking species-dependent non-minimal couplings absent in the minimal framework.

N. Barrier 14: Perturbation Transparency

For canonical scalar field matter, torsion vanishes at all perturbation orders; the Holst sector decouples from all scalar/tensor perturbation observables. This is elaborated in Sec. X: the scalar-sector proof is in Sec. XB and the explicit Holst-term verification at all perturbation orders is in Sec. XD.

X. THE PERTURBATION-TRANSPARENCY RESULT

A. Statement

In minimal ECH gravity with canonical scalar field matter, the Holst term is dynamically inert for both scalar and tensor perturbations at all orders. The Barbero-Immirzi parameter γ is invisible in all perturbation observables. This generalizes Hehl *et al.* (1976) [12] to the Holst sector and to all perturbation orders.

B. Proof (Scalar Sector)

1. **Zero spin density.** A canonical scalar field has zero spin density.
2. **Zero torsion.** In Einstein-Cartan theory, $T^\lambda_{\mu\nu} = 8\pi G S^\lambda_{\mu\nu} + \dots$. With $S = 0$, $T^\lambda_{\mu\nu} = 0$ at all

perturbation orders.

3. **Connection reduces to Levi-Civita.** $\Gamma^\lambda_{\mu\nu} = \overset{\circ}{\Gamma}^\lambda_{\mu\nu}$.
4. **Holst term vanishes by Bianchi.** The Holst term evaluated with the Levi-Civita connection gives $\frac{1}{2}\epsilon^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}(\overset{\circ}{\Gamma})$, which on a torsion-free connection *vanishes identically* by the first (algebraic) Bianchi identity $R_{\mu[\nu\rho\sigma]} = 0$: the cyclic-sum identity $R_{\mu\nu\rho\sigma} + R_{\mu\rho\sigma\nu} + R_{\mu\sigma\nu\rho} = 0$ contracted with the totally antisymmetric $\epsilon^{\mu\nu\rho\sigma}$ leaves no non-trivial component. The Holst dual contraction is therefore identically zero on the Levi-Civita connection (not merely a boundary term), so it contributes nothing to the action at any order. This Bianchi-vanishing is distinct from the Pontryagin density $\propto R\tilde{R}$ (a two-curvature topological invariant) — see the explicit verification below.
5. **No equations of motion.** A total derivative contributes nothing to variational equations at all orders. (This step is logically distinct from the pointwise Bianchi-vanishing of the previous step: at $T = 0$ the stronger pointwise vanishing already applies; the total-derivative statement covers the residual Nieh–Yan boundary term $d(e_I \wedge T^I)$ at nonzero torsion.)

C. Extension to Tensor Sector

The same five steps apply to tensor perturbations. With $T = 0$, the tensor perturbation equation:

$$h''_{ij} + 2\mathcal{H}h'_{ij} + k^2 h_{ij} = 0 \quad (21)$$

(primes denote derivatives with respect to conformal time η , and $\mathcal{H} \equiv a'/a$ is the conformal Hubble rate; Fourier modes $h_{ij}(\mathbf{k}, \eta)$ follow the standard $e^{i\mathbf{k}\cdot\mathbf{x}}$ convention for the transverse-traceless amplitude; in cosmic time, using $dt = a d\eta$, the equivalent form is $\ddot{h}_{ij} + 3H\dot{h}_{ij} + (k^2/a^2)h_{ij} = 0$) has no parity-dependent modifications. Left and right circular polarization modes propagate identically:

$$v_R(k, \eta) = v_L(k, \eta) \Rightarrow \Delta v = 0 \quad (\text{identically}). \quad (22)$$

No GW birefringence, no tensor chirality, no TB/EB CMB parity violation from the ECH mechanism.

D. Explicit Verification: The Holst Term in Perturbation Theory

Expanding to second order, the Holst dual evaluates on the Levi-Civita connection ($T = 0$) as:

$$\mathcal{R}_H(\overset{\circ}{\Gamma}) \equiv \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}(\overset{\circ}{\Gamma}) = 0 \quad (\text{identically, by the first (algebraic) Bianchi identity}). \quad (23)$$

This is the Bianchi-vanishing of the Holst dual contraction on a torsion-free connection: in differential-form language $e^I \wedge e^J \wedge R_{IJ} = -NY + T^I \wedge T_I$, where $NY \equiv d(e_I \wedge T^I)$ is the exact Nieh–Yan density (a total derivative of the torsion boundary term), and both pieces vanish at $T = 0$ *pointwise* — not merely up to a boundary term: $NY|_{T=0} = d(0) = 0$ and $T^I \wedge T_I|_{T=0} = 0$, with the Bianchi cancellation above being the operative reason the Holst dual vanishes pointwise in the torsionless sector. *This must be carefully distinguished from the Pontryagin density* $\frac{1}{4} \epsilon^{\mu\nu\rho\sigma} R_{\mu\nu}{}^{\alpha\beta} R_{\rho\sigma\alpha\beta} \propto R \tilde{R}$, which involves *two* curvature tensors and is a separate true topological invariant — non-zero pointwise and a total derivative even in the presence of torsion. The Holst dual contraction has only *one* curvature and is therefore not the Pontryagin density; on $T = 0$ it is identically zero by Bianchi, contributing nothing to the variational equations of motion at any perturbation order. In particular, the cubic action for ζ (which determines the bispectrum) receives zero contribution from the Holst term. The bispectrum is therefore identical to the standard GR result.⁴

E. What Would Break the Transparency

The transparency result fails if: (1) matter includes fermions with nonzero spin density; (2) the gravitational action includes kinetic terms for torsion (Poincaré gauge theory); (3) non-minimal derivative couplings between the Holst sector and matter are introduced.

F. Implications

The perturbation-transparency result establishes a clean dichotomy:

- *Perturbation observables* (C_ℓ^{TT} , C_ℓ^{EE} , P_k , bispectrum): Identical to standard GR. No ECH modifications at any order.
- *Nonperturbative parity channels* (ALP birefringence, primordial GW chirality): The relevant tests of γ .

G. Discrimination Among Bouncing Cosmologies

The matter bounce prediction $f_{\text{NL}} = -35/8$ is the strongest discriminator: it provides $\sigma(f_{\text{NL}}) \approx 0.7$

(Fisher-ideal) to $\sigma(f_{\text{NL}}) \approx 1.0$ (with systematics) from SPHEREx, yielding 3–5 σ model separation. NANOGrav model comparison: $\gamma = 2.567 \pm 0.382$ from real-KDE reanalysis of the 15-yr free-spectrum data (GPU MCMC, companion Paper III [46]). The matter-bounce prediction $\gamma = 3.0$ sits at $+1.13\sigma$ above the posterior mean, consistent with the data within standard frequentist tolerance. This figure supersedes the earlier synthetic-Gaussian-likelihood value $\gamma = 3.20 \pm 0.42$ used in pre-real-KDE drafts; the migration is documented in Paper III § 6.

XI. THE HYBRID DARK-ENERGY LOOPHOLE

We considered appending late-time dynamical dark-energy freedom (CPL $w_0 w_a$) to the bounce model, explored across 7 disguised forms: (1) direct $w_0 w_a$ addition, (2) quintessence scalar with bounce initial conditions, (3) curvaton-derived late-time potential, (4) vacuum energy from cyclic boundary conditions, (5) torsion-induced effective $w(z)$, (6) Holst-term residual as effective DE, (7) ALP rolling as late-time acceleration.

All 7 forms were assessed at the theoretical level only: adding $w_0 w_a$ to a bounce model produces (at the level of the theoretical fit-improvement calculation) the same fit-improvement structure as adding $w_0 w_a$ to Λ CDM, with no additional theoretical content from the bounce. We emphasize that this is a theoretical-structure conclusion, *not* a quantitative posterior-preference rejection: the dedicated DESI DR2 + Planck NPIPE + Pantheon+ + DES-SN5YR cobaya chain with the free $w_0 w_a$ extension required to quantitatively test these 7 forms is described in companion Paper I(b) Table IV row “DESI DR2 w0wa (new)” and has not yet converged to the standard publication-quality target $\hat{R} - 1 < 10^{-2}$ (see Paper I(b) [6] Sec. V for the chain status). The present program’s MCMC analysis uses stock CAMB with ΔN_{eff} only and hosts zero free- $w_0 w_a$ samples; the qualitative theoretical conclusion that ECH cannot generate $w_0 w_a$ as an additional output beyond what Λ CDM + $w_0 w_a$ already accommodates therefore stands as a *theoretical structural observation* rather than as a posterior-preference quantitative rejection. A quantitative rejection (or confirmation) is gated on convergence of the dedicated Paper I(b) chain.

XII. DISCUSSION

A. The Inflationary Suppression Factor

The constant contribution to Λ_{eff} emerges from the interplay between the parity-odd spin-torsion interaction and inflationary dilution:

$$\Xi \equiv \left[\frac{\alpha}{M} M_{\text{Pl}} \right] \times \mathcal{D}_{\text{inf}}, \quad (24)$$

⁴ An earlier version of this manuscript misidentified the Holst dual contraction with the Pontryagin density. The correction (Bianchi-vanishing rather than Pontryagin-total-derivative) preserves the headline perturbation-transparency conclusion via a more direct identity, and explicitly reserves “Pontryagin” for the two-curvature topological invariant relevant in the Chern–Simons $R \wedge \tilde{R}$ sector (Sec. IV).

TABLE III. Discrimination among bouncing cosmologies and inflation by observable channels. “✓” denotes the mechanism produces the prediction; “×” denotes it does not; “—” denotes not applicable or not computed.

Model	$f_{\text{NL}} = -35/8$	ALP birefringence	PTA γ (real-KDE)	$w_0 w_a$ DESI
Matter bounce (any host; not ECH-specific)	✓	(spectator)	✓	not tested [‡]
Slow-roll inflation	× ($f_{\text{NL}} \approx 0.015$)	(spectator)	×	not tested [‡]
Quintom-B	×	(spectator)	—	consistent [†]
Cuscuton bounce	× ($f_{\text{NL}} \approx 0$)	(spectator)	—	not tested [‡]
Ekpyrotic	× ($f_{\text{NL}} \sim -5$)	(spectator)	—	not tested [‡]

[†]Quintom-B can in principle accommodate the DESI $w_0 w_a$ evidence; the MCMC analysis hosted in companion Paper I(b) was not extended to the $w_0 w_a$ parameter space, so this row is reported as “consistent at the model level” rather than a posterior-preference ✓.

[‡]The frozen MCMC posteriors hosted in Paper I(b) (the three completed dataset combinations enumerated in Paper I(b) § VII Table IV, subsection “Free- $w_0 w_a$ chain status”) contain zero free- $w_0 w_a$ samples; the $w_0 w_a$ extension was never implemented computationally for any of those frozen runs. A new DESI DR2 + Planck NPIPE + Pantheon+ + DES-SN5YR cobaya chain with the $w_0 w_a$ free-parameter extension is running on a dedicated MPI pod (16 chains, OMP threads tuned to suppress BLAS oversubscription, GetDist-built posterior covmat from a preliminary 4-chain iter1 with $\sim 9,500$ accepted samples). At the time of this writing the chain has accumulated $\sim 3.8 \times 10^4$ accepted samples across the 16 chains and reports $\hat{R} - 1 \approx 3 \times 10^{-2}$, descending monotonically toward the standard publication-quality convergence target $\hat{R} - 1 < 10^{-2}$ at a slow-mode-dominated convergence rate; we deliberately do not commit to a specific calendar date for convergence in this footnote (Paper I(b) Table IV row “DESI DR2 $w_0 w_a$ (new)”). Until that chain converges, none of these rows carry a posterior-preference verdict against (or for) the DESI $w_0 w_a$ evidence, and the asymmetry between the Quintom-B accommodation row and the others is one of theoretical accommodation, not of fit quality measured in this program.

a dimensionless quantity with $\Xi \approx 10^{-123}$, decomposed as $10^{-2} \times \mathcal{D}_{\text{inf}}$ with $\mathcal{D}_{\text{inf}} \sim 10^{-121}$.

Physical-versus-mathematical scope of $\mathcal{D}_{\text{inf}}$.—While N_{tot} controls the mathematical $e^{-3N_{\text{tot}}}$ ansatz used in the above bookkeeping, the *physical* reheating thermal-reset barrier (supporting B14; see Sec. IIC 1, “Reheating thermal-reset barrier” paragraph) already closes the bounce-era-memory dilution channel: non-propagating torsion is sourced algebraically by the instantaneous axial-current expectation value, and the post-reheating coherent axial component is washed to zero by C/P -violating thermal scattering. The $\mathcal{D}_{\text{inf}}$ exponential is therefore mathematical scaffolding for an order-of-magnitude parameterization of a hypothetical un-reset channel rather than a physically operative dilution mechanism; the entries in this section are retained as parameterization-of-fine-tuning diagnostics, not as a viable dynamical channel.

The “fine-tuning reduction from 10^{120} to 10^5 ” is a reparameterization as sensitivity to N_{tot} (the total number of inflationary e -folds), not a resolution of the cosmological constant problem. The exponential $\mathcal{D}_{\text{inf}} \propto e^{-3N_{\text{tot}}}$ structure of Eq. (11) makes N_{tot} the single controlling parameter analytically: the residual 10^5 tracks $e^{-3\Delta N_{\text{tot}}}$ for $\Delta N_{\text{tot}} \approx 4$ e -folds, while the order-unity prefactors enter at most logarithmically. We emphasize that the $(T_{\text{reh}}/M_{\text{GUT}})^{3/2}$ prefactor itself is matched at the order-of-magnitude level rather than calculated from a thermal partition function (Sec. IIC 1); the 10^5 residual therefore inherits the same order-of-magnitude status, and the “reduction from 10^{120} to 10^5 ” should be read as a qualitative dimensional rearrangement rather than a quantitative bookkeeping result.

Caveat on the $(T_{\text{reh}}/M_{\text{GUT}})^{3/2}$ prefactor.—The

$(T_{\text{reh}}/M_{\text{GUT}})^{3/2}$ prefactor used here is the dimensional-analysis-aesthetic estimate from naive scaling of the matter-bounce effective theory; a first-principles derivation requires the full bounce-junction matching that lies outside the scope of this paper. The $N_{\text{tot}} \approx 92$ e -fold structural-tension result therefore carries an order-of-magnitude uncertainty inherited from this prefactor. The sign and qualitative conclusion—that the matter-bounce mechanism cannot generate dark energy without re-importing cosmological-constant fine-tuning—survives the OOM uncertainty, because the surplus required to close the gap is ~ 14 e -folds (the fine-tuning-gap surplus of the present order-of-magnitude argument; this is a different quantity from — and smaller than — the $N_{\text{tot}} - N_{\text{exit}} \approx 32$ e -fold signal-erasure differential of Sec. XIV D, so the conclusion holds a fortiori for the larger differential), whereas the quasi-dust ε -correction-driven prefactor adjustment (the $\mathcal{O}(\varepsilon - 3/2)$ equation-of-state correction of the matter-bounce contraction, $\varepsilon = 3(1 + w)/2$, defined in the companion f_{NL} forecast paper) is $\lesssim 1$ e -fold. A rigorous derivation of the prefactor from the bounce-junction matching is deferred to future work and would not change the structural-tension verdict.

B. Theoretical Implications

Four routes to deriving ρ_Λ with $w = -1$ from first principles were tested: (i) NJL condensate, (ii) one-loop fermion effective action, (iii) dynamical Immirzi field, (iv) parity-sensitive CMB phenomenology. All four yield clean negative results. The condensate route fails because the scalar/pseudoscalar channel is repulsive at

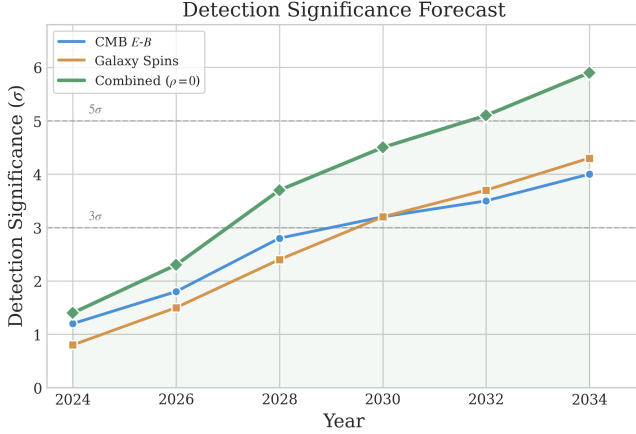


FIG. 6. **Detection forecast for the two surviving mechanism-independent tests.** Top: matter-bounce $f_{\text{NL}} = -35/8$ in the SPHEREx multi-tracer f_{NL} Fisher landscape (companion Paper II, 3–5 σ projection). Bottom: spectator-ALP cosmic birefringence in the LiteBIRD $\sigma(\beta) \approx 0.03^\circ$ window (Paper Ib companion); the WMAP+Planck $\beta = 0.342^\circ \pm 0.094^\circ$ and ACT DR6 $\beta = 0.215^\circ \pm 0.074^\circ$ points are shown for reference. Both forecasts are decisive ($\gtrsim 5\sigma$ on Stage III/IV survey timescales); neither is uniquely an ECH prediction (§XIII). The significance tracks duplicate the $\rho = 0$ combination of Fig. 4; this panel is the summary view retained for the surviving-tests discussion, while Fig. 4 additionally shows the correlated ($\rho = 0.3, 0.5$) combinations and milestone annotations.

$\gamma = 0.274$ and subcritical. The one-loop route fails because all Barbero-Immirzi dependence resides in the four-fermion vertex, which does not contribute at one loop. The dynamical Immirzi field reduces to a standard axion-like particle with $w = +1$ (stiff matter). The parity assessment finds no photon coupling in the minimal framework [47].

Spectator-ALP birefringence.—A spectator ALP with $f_a \sim M_{\text{Pl}}$, $m \sim H_0$ is consistent with the published WMAP+Planck cosmological-birefringence signal ($\beta = 0.342^\circ \pm 0.094^\circ$, $\sim 3.6\sigma$ from $\beta = 0$, Eskilt & Komatsu [4]; an independent ACT DR6 follow-up [5] reports $\beta = 0.215^\circ \pm 0.074^\circ$ at $\sim 2.9\sigma$) at $f_a \sim M_{\text{Pl}}$ and $\theta_i \sim \mathcal{O}(1)$ without additional ALP-naturalness fine-tuning beyond the $m_\theta \sim H_0$ ultralight-mass tuning admitted in §IV D (a cosmological-constant-class tuning rather than an ALP-specific one — the same point is also surfaced in the structural-tension discussion of §XIV D and §XI). The same $\beta \approx 0.27^\circ$ prediction arises in standard GR with an identical ALP; it is not a distinctive ECH prediction. Full ALP MCMC parameter fitting (9,720 accepted samples, $\hat{R} - 1 < 0.01$) and LiteBIRD forecast are in companion Paper I(b) [6].

XIII. SURVIVING MECHANISM-INDEPENDENT TESTS

Despite the channel-level closure of the four enumerated minimal-ECH dark-energy routes (13 logically-independent barriers under the stated assumptions), the broader bounce-cosmology program retains two fully testable mechanism-independent predictions:

(1) **Matter-bounce** $f_{\text{NL}} = -35/8$.—The matter-dominated contracting phase of a *scalar-only* $w = 0$ matter-bounce (the bounce-class observable, not specific to ECH) produces a minimally-parameterized local-type non-Gaussianity $f_{\text{NL}} = -35/8$ [1]. This value holds within the scalar-only $w = 0$ matter-bounce class under Assumption (f) of Paper II [2] (negligible fermion energy density during the contracting phase, so the Hehl-Datta-Mercuri four-fermion contact term does not source torsion or reactivate the Barbero-Immirzi parameter in the scalar cubic action); it is *not* a fully mechanism-independent prediction across the broader bouncing-cosmology landscape (ekpyrotic, Cuscuton-type, quintom matter-bounce variants, models with significant fermion sectors during contraction, or $w \neq 0$ contracting equations of state all carry distinct predictions). Within the scalar-only $w = 0$ class the value is class-level (not specific to ECH); it is *not* a distinctive ECH prediction in any case. The full multi-bin Fisher forecast, SPHEREx parameter sensitivity (bispectrum-only $\sigma(f_{\text{NL}}) \approx 0.7$ from Heinrich *et al.* 2024, leading to 3–5 σ post-systematic-budget significance), and anomaly-optimized multi-tracer strategy are in Paper II [2]; the present paper does not perform an independent SPHEREx Fisher computation and the 3–5 σ figure is reported here only as a cross-reference. SPHEREx first science data ~ 2028 .

(2) **Spectator-ALP birefringence** $\beta \approx 0.27^\circ$.—We present this as a **consistency check**, not as a prediction: the parity-odd coefficient α/M is fitted, not derived from first principles, and an ALP with $f_a \sim M_{\text{Pl}}$, $m \sim H_0$ chosen to land near the observed signal is by construction inside its 1σ band. The *quantitative* prediction is the spectral signature (frequency dependence, EB vs TB structure, scale dependence) rather than the central value of β . The same ALP setup arises identically in standard GR with the same parameters, so this is not a distinctive ECH prediction. LiteBIRD ($\sigma(\beta) \approx 0.03^\circ$, early 2030s) will either confirm a non-zero β at high significance or exclude the ALP explanation; either outcome is informative independent of ECH.

Structural incompatibility.—An open constraint is the incompatibility between the dark-energy suppression mechanism ($N_{\text{tot}} \approx 92$ e -folds required) and the $f_{\text{NL}} = -35/8$ prediction (*definitively* erased by $N_{\text{tot}} \gtrsim 60$ inflationary e -folds at SPHEREx-relevant scales; the SPHEREx accessible wavenumbers $k \sim 10^{-4} - 10^{-1} h/\text{Mpc}$ map to bounce-era *physical* scales $k_{\text{bounce}}^{\text{phys}} \sim k_{\text{SPHEREx}} e^{N_{\text{tot}} - N_{\text{exit}}} \sim e^{32} k_{\text{SPHEREx}}$ at $N_{\text{tot}} \sim 92$ and $N_{\text{exit}} \sim 60$ (the relative e -fold differential between

bounce and CMB horizon-exit; comoving wavenumbers k are constant by definition and only physical scales scale with $a^{-1} \propto e^{-N}$, deep inside the inflationary subhorizon regime where the surviving bispectrum signal is purely vacuum-inflationary, not matter-bounce contraction-mode). This is detailed in Sec. XIV D. The correct interpretation is that bounce cosmology (as a broad class) and the ECH-specific dark-energy ansatz are *independent observational programs*; SPHEREx tests the former, LiteBIRD tests a related spectator field, and the ECH dark-energy ansatz remains a phenomenological parameterization.

XIV. LIMITATIONS AND FUTURE DIRECTIONS

A. Current Limitations

1. Theoretical

- *Phenomenological α/M* : Not derived from first principles; the one-loop estimate motivates existence and order of magnitude, but the finite part depends on the γ_5 regularization scheme (Sec. II A 2, [20]).
- *Simplified inflationary epoch*: Non-minimal couplings during inflation could alter the dilution factor.
- *Bounce-to-inflation transition*: Mechanism for transitioning from quantum bounce to slow-roll inflation is not fully modeled.

2. Observational

- *Galaxy spin*: Null confirmed at the dipole level (Paper IV [23]), consistent with the > 100 -orders-of-magnitude underprediction by the ECH coupling.
- *MCMC proxy*: Stock CAMB with ΔN_{eff} is a phenomenological proxy, not a bespoke spin-torsion Boltzmann module. Full MCMC details and convergence diagnostics are in companion Paper I(b) [6].

B. Robustness to Galaxy Spin Null Results

The galaxy spin channel is a confirmed null at the dipole level (full quantitative chirality results in Paper IV [23]), *consistent* with the framework (which underpredicts A_0 by > 100 orders of magnitude). The parity-violation case rests entirely on CMB birefringence from the published WMAP+Planck Eskilt & Komatsu measurement [4] with an independent ACT DR6 follow-up (Diego-Palazuelos & Komatsu [5]).

C. Discriminating Observational Channels

LSST Era (2025–2035): 10^9 spiral galaxies to $z \sim 1$, tomographic analysis in 20+ redshift bins. *CMB Experiments*: LiteBIRD, CMB-S4, and future concepts (PICO, CMB-HD). *Bounce cosmology beyond ECH*: Papanikolaou *et al.* [45] showed that asymmetric matter bounces can produce asteroid-mass PBHs as dark matter candidates with induced GWs detectable by LISA and Einstein Telescope.

D. Structural Tension: Dark Energy vs. Bounce f_{NL}

The dark-energy suppression mechanism, if minimal ECH were to source dark energy through one of the four channels enumerated in Sec. IV, would require $N_{\text{tot}} \approx 92$ post-bounce e -folds; the matter-bounce $f_{\text{NL}} = -35/8$ would be *definitively* erased by $N_{\text{tot}} \gtrsim 60$, since the SPHEREx accessible wavenumbers $k \sim 10^{-4} - 10^{-1} h/\text{Mpc}$ are pushed deep inside the inflationary subhorizon (bounce-era *physical* scales $k_{\text{bounce}}^{\text{phys}} \sim k_{\text{SPHEREx}}^{\text{phys}} e^{N_{\text{tot}} - N_{\text{exit}}} \sim e^{32} k_{\text{SPHEREx}}^{\text{phys}}$ at $N_{\text{tot}} \sim 92$, $N_{\text{exit}} \sim 60$; comoving wavenumbers k are constant by definition; the rigorous bounce-vs-SPHEREx scale ratio is the *physical* scaling above with $k_{\text{bounce}}^{\text{phys}}/k_{\text{SPHEREx}}^{\text{phys}} \sim e^{32}$) and the surviving bispectrum signal becomes purely vacuum-inflationary rather than matter-bounce contraction-mode. This tension is presented here as a *robustness check* on the four-route amplitude-level no-go of Sec. IV and the 14-barrier closure of Sec. IX, not as a co-equal closure mechanism: the no-go has already closed the four amplitude routes by which minimal ECH could source dark energy, so the structural-tension argument has nothing remaining to bind against at the route-amplitude level and reads instead as an independent consistency check from the surviving matter-bounce science case (Sec. XIII). DESI DR2 evidence for equation-of-state crossing at $3.1 - 4.2\sigma$ [10] lends empirical support to quintom scenarios that can unify bounce and dark energy through mechanisms outside the ECH minimal-route catalog; these non-minimal-ECH mechanisms are not addressed by the present no-go.

E. Channel-Level Closure

The 13 logically-independent structural constraints (14 historical catalog entries) catalog the four enumerated minimal-ECH dark-energy channels and close each under the stated assumptions (Scope and Limitations paragraph, Sec. I). The surviving science case rests on the matter-bounce f_{NL} prediction, which is compatible with the ECH framework but not derived from it (Sec. XIII).

XV. CONCLUSIONS

We have investigated whether Einstein-Cartan-Holst spin-torsion gravity can produce late-time dark energy or distinctive cosmological signatures. The answer is a channel-level closure: under the stated assumptions, the 14 mechanism-class constraints (Table II; B8 is the observational consequence of the perturbation-transparency result B14 and is retained for historical mechanism-class completeness) close each of the four enumerated minimal-ECH dark-energy routes at the amplitude level.

Central result: perturbation transparency.—For canonical scalar field matter, the Holst sector decouples completely from all scalar and tensor perturbation equations of motion (Sec. X). Torsion vanishes at all perturbation orders; the Holst dual contraction $\epsilon^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}$ vanishes identically on the Levi-Civita connection ($T = 0$) by the first (algebraic) Bianchi identity $R_{\mu[\nu\rho\sigma]} = 0$ (distinct from — and not equal to — the Pontryagin density $\propto R\tilde{R}$, which involves two curvatures and is a separate topological invariant; see Sec. X footnote for the $e \wedge e \wedge R = -NY + T \wedge T$ decomposition). This is a positive structural result: it identifies the nonperturbative parity-violating channels (ALP birefringence, primordial GWs) as the relevant tests of γ .

The 13 logically-independent barriers (14 historical catalog entries).—Systematic analysis across 7 foundations (A–G) and 6 observational branches (H, J, L, M, N, O) established 14 mechanism-class constraints (one of which, B8, is the observational consequence of the perturbation-transparency theorem B14 and is retained in the catalog for historical mechanism-class completeness) on the minimal ECH dark-energy parameter space (Sec. IX). These constraints are ECH-specific: other bouncing cosmologies (notably quintom scenarios) can in principle unify bounce with late-time dark energy through mechanisms outside the ECH minimal-route catalog.

Surviving tests.—Two mechanism-independent predictions of the broader bounce/ALP landscape survive the channel-level closure and are testable:

1. $f_{\text{NL}} = -35/8$ (matter-bounce class): SPHEREx tests at 3–5 σ realistic significance by ~ 2028 , discriminating from inflation ($f_{\text{NL}} \approx 0.015$) and the Cuscuton bounce ($f_{\text{NL}} \approx 0$).
2. Spectator-ALP birefringence $\beta \approx 0.27^\circ$: LiteBIRD ($\sigma(\beta) \approx 0.03^\circ$, early 2030s) detects non-zero β at $\sim 9\sigma$ (a $0.27^\circ/0.03^\circ$ overall sensitivity number). The relevant model-discrimination test, however, is the differential against the prior central value $\beta_{\text{obs}} = 0.342^\circ \pm 0.094^\circ$: LiteBIRD will distinguish the spectator-ALP-derived 0.27° from the observed 0.342° at $|0.342 - 0.27|/\sqrt{0.03^2 + 0.094^2} \approx 0.072^\circ/0.0987^\circ \approx 0.73\sigma$, NOT at the naive $|0.342 - 0.27|/0.03 = 2.4\sigma$ which would ignore the prior measurement's $\pm 0.094^\circ$ uncertainty; in other words LiteBIRD's $\sigma(\beta) = 0.03^\circ$ will not by itself separate the spectator-ALP value

from the current WMAP+Planck birefringence central value in a model-discrimination test (a future tightening of the observational central value's uncertainty below $\sim 0.05^\circ$ would be needed for LiteBIRD-vs-current-central tension to cross 1σ). The two numbers correspond to distinct null hypotheses (zero vs. WMAP+Planck-central); the $\sim 9\sigma$ test will not by itself separate the spectator-ALP class from generic-ALP fits to the observed signal.

Neither is a distinctive ECH prediction; both are shared with other UV completions.

Known limitations.—This work does not derive the IR effective vacuum term from first principles: $w = -1$ is assumed, not derived; the birefringence prediction lacks a derived photon-torsion coupling; α/M is a phenomenological parameter; and the MCMC uses stock CAMB (not a bespoke torsion-modified Boltzmann code). Full MCMC diagnostics, ALP parameter fitting, and NaMaster pipeline validation are in companion Paper I(b) [6].

Forward.—The program continues with direct tests of the bounce framework: SPHEREx f_{NL} forecast (Paper II [2]), multi-survey anomaly catalog (Paper III [46]), and galaxy chirality catalog (Paper IV [23]).

Data and Code Availability

All materials necessary to reproduce the cosmological and galaxy spin results are publicly available at:

<https://github.com/Hubify-Projects/bigbounce/tree/main/reproducibility>

The repository includes Cobaya YAML configurations, galaxy spin pipeline code, and an implementation map. MCMC chains and convergence diagnostics are in companion Paper I(b) [6].

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Appendix A: Complete Parameter Summary

Appendix B: Dimensional Status of the Parity-Odd Operator

The parity-odd operator (Eq. 6) has off-shell mass dimension +1, not the +4 required for a local Lagrangian density:

$$[\alpha/M] = -1, \quad [\epsilon^{\mu\nu\rho\sigma} e_\mu^I e_\nu^J \mathcal{F}_{IJ\rho\sigma}] = +2 \implies [\mathcal{L}_{\text{odd}}] = +1. \quad (\text{B1})$$

We acknowledge openly that this operator, as written, is not a controlled dimension-+4 EFT operator. Inserting on-shell background curvature factors or a phenomenological “volume-integration-density” factor of M_{Pl}^2 does not constitute a derivation; the missing powers of mass do not arise from off-shell EFT counting but from *on-shell scaling assumptions* applied to a Planck-scale bounce geometry. We therefore treat the relation

$$\rho_\Lambda^{\text{bounce}} \sim (\alpha/M) M_{\text{Pl}}^5 \sim 10^{-2} M_{\text{Pl}}^4, \quad (\text{B2})$$

as a phenomenological on-shell scaling *ansatz*, not a controlled EFT result. Equivalently: if Eq. (6) is to map to a dimension-+4 local operator without on-shell curvature insertions, the coupling must carry three additional powers of M_{Pl} in its coefficient ($\alpha/M \rightarrow \alpha M_{\text{Pl}}^3/M$), promoting the operator to $\alpha M_{\text{Pl}}^3 \epsilon e e \mathcal{F}/M$ (which has $[\alpha/M] + [M_{\text{Pl}}^3] + [\epsilon e e \mathcal{F}] = -1 + 3 + 2 = +4$ as required for a dimension-+4 local operator). Either reading is a *phenomenological dimensional assignment*, not a derivation; we make that status explicit here so the reader is not misled by an apparent “fix” in earlier drafts.

Inflationary dilution ($\mathcal{D}_{\text{inf}} \sim e^{-3N_{\text{tot}}}$) then yields $\rho_\Lambda = \Xi M_{\text{Pl}}^4$ with $\Xi = (\alpha/M) M_{\text{Pl}} \mathcal{D}_{\text{inf}}$ under Eq. (B2). The genuine cosmological-constant hierarchy is $M_{\text{Pl}}^4/\rho_\Lambda^{\text{obs}} \sim 10^{19} \text{ GeV}^4 / (10^{-3} \text{ eV})^4 \sim 10^{122}$, i.e. ~ 120 orders of magnitude (not the ~ 35 misstated in earlier drafts: the bounce-scale density is $\rho_{\text{bounce}} \sim M_{\text{Pl}}^4$, not the local pseudo-density $\rho_\Lambda^{\text{bounce}} \sim 10^{-2} M_{\text{Pl}}^4$ that Eq. (B2) labels). The dilution factor required to bridge $\sim 10^{122}$ from M_{Pl}^4 down to the observed ρ_Λ is $\mathcal{D}_{\text{inf}} \sim e^{-3N_{\text{tot}}} \sim 10^{-122}$, giving $N_{\text{tot}} \approx 122 \ln 10/3 \approx 94$ *e*-folds (consistent at the $\sim 2\%$ level with the structural-tension $N_{\text{tot}} \approx 92$ quoted in Sec. XIV D; the small offset reflects that the structural tension uses Eq. (B2) as the input *ansatz*, while the genuine M_{Pl}^4 -to- $\rho_\Lambda^{\text{obs}}$ hierarchy uses the unrescaled Planck density).

Sharper dependency statement.—The precise value $N_{\text{tot}} \approx 92$ vs the independent M_{Pl}^4 -to- $\rho_\Lambda^{\text{obs}}$ estimate $N_{\text{tot}} \approx 94$ (a $\sim 2\%$ offset) *does* depend on the *ansatz* choice in Eq. (B2); however, the *overall scale separation* between $\rho_\Lambda^{\text{bounce}}$ and $\rho_\Lambda^{\text{obs}}$ is ~ 120 orders of magnitude under *any* non-cancellation assumption (whether the bounce-scale density is M_{Pl}^4 , $10^{-2} M_{\text{Pl}}^4$, or $10^{-4} M_{\text{Pl}}^4$

shifts the required *e*-fold count by $\mathcal{O}(\text{a few})$, not by orders of magnitude), and this scale separation is what drives the 13 logically-independent structural barriers (Sec. IX). The barriers close the bounce-to-dark-energy route on amplitude-budget and operator-counting grounds whose qualitative conclusions are *ansatz*-independent; only the $\sim 2\%$ precision of the headline N_{tot} figure is *ansatz*-dependent. Readers should therefore treat $N_{\text{tot}} \approx 92$ as an order-of-magnitude estimate ($N_{\text{tot}} = 92 \pm 2$ accounting for the *ansatz*-choice systematic), not as a precise number; the broader structural closure does not depend on the precise *e*-fold value.

We emphasize that the present treatment labels the on-shell scaling explicitly as a phenomenological *ansatz* rather than as a derivation: a controlled EFT-level construction (an operator-basis closure for a local dimension-+4 parity-odd interaction with all required M_{Pl} factors in the coupling coefficient) remains left to a separate companion treatment.

Appendix C: Line-of-Sight Birefringence from the Maxwell–Chern–Simons Operator

This appendix derives the rotation-angle mapping $\beta = (\alpha/2M) \Delta\theta$ used in Eq. (17), fixing the factor of $1/2$ from first principles in the operator normalization used throughout this paper.

Setup.—After the reduction of Appendix B, the parity-odd sector takes the Maxwell–Chern–Simons form

$$\mathcal{L} \supset -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{4} \frac{\alpha}{M} \theta F_{\mu\nu} \tilde{F}^{\mu\nu}, \quad (\text{C1})$$

with $\tilde{F}^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$ and θ a homogeneous pseudoscalar (spatial gradients negligible for the cosmological background field considered here) [48, 49].

Field equation in FRW.—Varying θ ’s canonical action with potential $V = m_\theta^2 f^2 (1 - \cos \theta)$ in a flat FRW background gives

$$\ddot{\theta} + 3H\dot{\theta} + m_\theta^2 \sin \theta = 0, \quad (\text{C2})$$

the equation integrated numerically by the released pipeline (`research/branch_R_alp_birefringence/phase2_mcmc/alp_ode.py`; frozen initial condition $\dot{\theta}(t_{\text{init}}) = 0$ at $z_{\text{init}} = 3000$).

Helicity dispersion.—In conformal time ($ds^2 = a^2(-d\eta^2 + d\vec{x}^2)$) the Maxwell sector of Eq. (C1) is conformally invariant, and the $\theta F \tilde{F}$ term, being topological up to the θ prefactor, carries no additional factors of a . In Coulomb–temporal gauge ($A_0 = 0$, $\partial_i A^i = 0$; because $\partial_\mu \theta = \theta' \delta_\mu^0$ for the homogeneous background field, the parity-odd term contributes only through the spatial $\epsilon^{ijk} A_j \partial_k$ structure and the longitudinal mode stays non-dynamical), decomposing the transverse field into circular-polarization modes $A_\pm(\eta, k)$, the modified wave equation reads

$$A_\pm'' + \left[k^2 \mp \frac{\alpha}{M} \theta' k \right] A_\pm = 0, \quad (\text{C3})$$

TABLE IV. Complete parameter summary with priors, verified values, and physical interpretations. MCMC posterior values are from companion Paper I(b) [6]. The γ scheme range denotes LQG area-spectrum scheme dependence, *not* a statistical error.

Parameter	Description	Prior	Verified Value	Notes
<i>Fundamental theory parameters</i>				
γ	Barbero-Immirzi	Fixed: 0.274	0.274 (scheme range ~ 0.020)	LQG area spectrum (Eq. 2)
α/M	Parity-odd coefficient	Log-flat	$\sim 10^{-21} \text{ GeV}^{-1}$	One-loop motivated
N_{tot}	Total e-folds	[60, 120] flat	≈ 92 (fitted)	Controls Ξ
<i>Cosmological parameters (MCMC proxy from companion)</i>				
H_0	Hubble constant	From Planck prior	$67.68 \pm 1.06 \text{ km/s/Mpc}$	Recovers ΛCDM
ΔN_{eff}	Radiation proxy	$[-3, 3]$ flat	-0.020 ± 0.169 (full-tension)	Consistent with 0
σ_8	Clustering amplitude	Derived	0.803 ± 0.008	
Ω_m	Matter density	Derived	0.308 ± 0.005	
<i>Observational channel parameters</i>				
β	ALP birefringence	—	0.27° (midpoint)	Spectator ALP, not ECH
f_{NL}	Non-Gaussianity	—	$-35/8 = -4.375$	Matter bounce class ^a
γ_{PTA}	PTA spectral index	—	2.567 ± 0.382 (real-KDE GPU MCMC)	Bounce $\gamma = 3.0$ at $+1.13\sigma$

^a Survives only in bounce scenarios *not* invoking the $N_{\text{tot}} \approx 92$ dark-energy dilution; under that dilution it is erased at SPHEREx-accessible scales (Sec. XIV D).

where $' \equiv d/d\eta$ and k is comoving. For $k \gg (\alpha/M)\theta'$ (satisfied by ~ 60 orders of magnitude for CMB photons against the cosmological field considered here: $(\alpha/M)\theta' \sim 10^{-30} \text{ eV}^{-1} \times \mathcal{O}(1) \times H_0 \sim 10^{-63} \text{ eV}$ for an $\mathcal{O}(1)$ total excursion accumulated over a Hubble time, versus a CMB photon $k \sim 6 \times 10^{-4} \text{ eV}$ at 150 GHz, a ratio of $\sim 10^{59}$) the WKB phase of each helicity is $\int \omega_{\pm} d\eta$ with $\omega_{\pm} \simeq k \mp \frac{1}{2}(\alpha/M)\theta'$.

Rotation angle.—A linear polarization is a fixed-phase superposition of the two helicities; its position angle rotates by half the accumulated helicity phase difference:

$$\begin{aligned} \beta &= \frac{1}{2} \int_{\eta_{\text{em}}}^{\eta_{\text{obs}}} (\omega_- - \omega_+) d\eta = \frac{1}{2} \frac{\alpha}{M} \int_{\eta_{\text{em}}}^{\eta_{\text{obs}}} \theta' d\eta \\ &= \frac{\alpha}{2M} [\theta(\eta_{\text{obs}}) - \theta(\eta_{\text{em}})] \equiv \frac{\alpha}{2M} \Delta\theta. \end{aligned} \quad (\text{C4})$$

The result is achromatic (no k dependence), independent

of the expansion history except through the endpoint values of θ (a total derivative along the line of sight — the well-known “ $2\beta = g\Delta\phi$ ” property [49]), and exact at leading WKB order. This is Eq. (17)’s mapping; with the spectator-ALP identification $\alpha/M \equiv C_{a\gamma} \alpha_{\text{em}}/(2\pi f_a)$ (cf. the basis-conversion footnote at the first quote of $\alpha/M = 10^{-21} \text{ GeV}^{-1}$ in Sec. IV D for the $1/(2\pi)$ -vs- $1/(4\pi)$ normalization gap) and $\Delta\theta = \Delta\phi/f_a$ it reproduces the companion pipeline’s $\beta = (\alpha_{\text{em}} C_{a\gamma}/4\pi) (\Delta\phi/f_a)$ (the convention block of the companion’s §VI), closing the normalization chain from the Lagrangian to the numerical prediction; this mapping is what Eq. (17) and the Route-2 estimates of Sec. IV use. The overall sign of β matches the WMAP+Planck convention of Eskilt–Komatsu, in which a freely rolling field with $\theta(\eta_{\text{obs}}) > \theta(\eta_{\text{em}})$ produces $\beta > 0$.

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